16/05/2026

Code-P

Aakash

Medical|IIT-JEE|Foundations

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MM : 720

Project Phoenix for Re-NEET-2026_PST-01P

Time : 180 Min.

Topics Covered:
Physics: Units & Measurements, Motion in a Straight Line, Motion in a Plane, Laws of Motion, Work, Energy and Power, System of Particles and Rotational Motion, Gravitation
Chemistry: Chemical Bonding and Molecular Structure, The p-Block Elements (Group-13 & 14), Equilibrium, Some Basic Concepts of Chemistry, Structure of Atom
Botany: Cell: The unit of life, Plant kingdom, Biological classification, Photosynthesis in higher plants, Respiration in plants
Zoology: Animal Kingdom, Structural Organization in Animals, Biomolecules, Breathing and Exchange of gases

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

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PHYSICS

1. How many significant figures are there in 427.003?
(1) 1 (2) 3
(3) 5 (4) 6

2. A substance of mass 42.60 g occupies 1.2 cm^3 of volume.
The density of substance (in g cm^{-3}) with correct number of significant figures is
(1) 36 (2) 35.5
(3) 35.0 (4) 35

3. In the following questions, a statement of assertion (A) is followed by a statement of reason (R).
A : The acceleration at a particular instant is the slope of the tangent to the *v-t* curve at that instant.
R : A change in velocity may involve either change in magnitude or change in direction or both.
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

4. A body is moving with constant speed. The acceleration of body
(1) May be varying
(2) May be zero
(3) May be of constant magnitude
(4) All of these

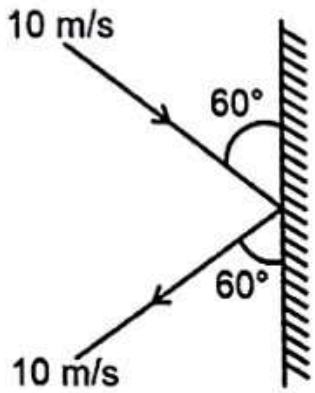
5. A person can swim in still water with a speed of 5 ms^{-1} in a river of width 200 m. He wishes to reach the exactly opposite point on the other bank. If the river is flowing at 3 ms^{-1} then the time taken to cross the river is
(1) 20 s (2) 100 s
(3) 50 s (4) 40 s

6. Speed of a projectile at the point of projection is $\sqrt{2}$ times the speed at its maximum height, the angle of projection of the projectile is
(1) 30° (2) 45°
(3) 60° (4) $\tan^{-1}(\sqrt{2})$

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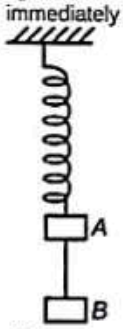
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7. A body of mass 3 kg hits a wall at an angle of 60° with the wall and returns at the same angle as shown in the figure. The impact time was 0.2 s. The average force exerted on the wall is



(1) $150\sqrt{3}$ N (2) $50\sqrt{3}$ N
(3) 100 N (4) $75\sqrt{3}$ N

8. Two blocks *A* and *B* of masses 2 *m* and *m* respectively are connected by a massless inextensible string. The whole system is suspended by a massless spring as shown in figure. The magnitude of acceleration of *A* and *B* immediately after the string is cut, are respectively



(1) g, g (2) $2g, g$
(3) $g, \frac{g}{2}$ (4) $\frac{g}{2}, g$

9. Two bodies of masses $3m$ and $4m$ have their kinetic energy in the ratio 3 : 1. The ratio of their momenta is
(1) 2 : 1 (2) 3 : 2
(3) 3 : 1 (4) 4 : 3

10. A force $\vec{F} = \left(x\hat{i} + y^2\hat{j}\right)$ N acts on a particle and the particle moves from (1, 2) m to (– 3, 4) m. Find the work done by the force $\vec{F}$.
(1) $\frac{28}{3}$ J (2) $\frac{38}{3}$ J
(3) $\frac{68}{3}$ J (4) $\frac{48}{3}$ J

11. The radius of gyration a uniform rod of mass *M* and length *l* about an axis passing through mid-point of the rod and perpendicular to the length will be
(1) $\frac{l}{\sqrt{3}}$ (2) $\frac{l}{2}$
(3) $\frac{l}{2\sqrt{3}}$ (4) $\frac{l}{3\sqrt{2}}$

12. Two bodies of masses 5 kg and 1 kg are moving with velocities $\left(2\hat{i} + 4\hat{j}\right)$ m/s and $6\hat{j}$ m/s respectively. The velocity of their COM is
(1) $\left(\frac{5\hat{i} + 13\hat{j}}{3}\right)$ m/s
(2) $\left(\frac{10\hat{i} + 6\hat{j}}{6}\right)$ m/s
(3) $\left(\frac{12\hat{i} + 6\hat{j}}{7}\right)$ m/s
(4) $\left(\frac{9\hat{i}}{5}\right)$ m/s

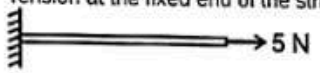
13. If a torque of 20 N m is applied on a rigid body of moment of inertia 5 kg m^2, then its angular acceleration will be
(1) 2 rad s^{-2} (2) 4 rad s^{-2}
(3) $\frac{1}{4}$ rad s^{-2} (4) $\frac{1}{2}$ rad s^{-2}

14. Kepler's law regarding constancy of areal velocity of a planet is the consequence of the law of conservation of
(1) Energy (2) Angular momentum
(3) Liner velocity (4) Linear momentum

15. **Statement I:** Weight experienced by astronaut in a spaceship while orbiting the earth is zero.
Statement II: Kinetic energy of a satellite is positive.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both statement I and statement II are correct
(2) Statement I is correct but statement II is incorrect
(3) Statement I is incorrect but statement II is correct
(4) Both statement I and statement II are incorrect

16. If masses of two spherical bodies are quadrupled and the distance between their centres is doubled, then percentage increase in force of gravitation between them will be
(1) 400% (2) 200%
(3) 100% (4) 300%

17. A massless string of length 20 m is fixed at one end and a force of 5 N is applied to other end as shown in figure. Tension at the fixed end of the string is



(1) Zero (2) 5 N
(3) $\frac{3}{4}$N (4) $\frac{1}{4}$N

18. The average power of a pump which can pump 200 kg of water to height 20 m in 20 s is
[g = 10 m/s^2]
(1) 2 kW (2) 20 kW
(3) 200 kW (4) 2000 kW

19. The 9th division of vernier scale coincides with 6th division of main scale in a vernier calliper. If one main scale division is 0.01 mm, then the least count of instrument is
(1) 0.01 mm (2) 0.001 mm
(3) 0.0033 mm (4) 0.0066 mm

20. Percentage errors in the measurement of momentum and mass are 3% and 2% respectively. The maximum error in the estimate of kinetic energy obtained by measuring mass and speed will be
(1) 2% (2) 6%
(3) 10% (4) 8%

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21. The value of 10 J in a system of units, which has 10 cm, 1 kg and 30 s as the units of length, mass and time respectively, is

(1) 2×10^4 units (2) 9×10^5 units
(3) 15×10^5 units (4) 18×10^5 units

22. If area (A), velocity (v) and density (ρ) are base units, then the dimensional formula of force can be represented as

(1) $[\rho A v^2]$ (2) $[\rho A v^3]$
(3) $[\rho A v]$ (4) $[\rho A^2 v^2]$

23. A train of length 300 m travelling at 30 m s^{-1} crosses another train of length 200 m travelling at 20 m s^{-1} in opposite direction. Time taken by first train to pass the second train is

(1) 10 s (2) 30 s
(3) 50 s (4) 2 s

24. A ball is dropped from a height 60 m. At the same time another ball is projected vertically upward with 30 m/s from ground. Before crossing each other distance travelled by balls are in ratio

(1) 1 : 1 (2) 1 : 2
(3) 1 : 3 (4) 1 : 4

25. A particle is moving in a circle of radius 4 m with velocity $v = 2t$ where t is time in second. Then the acceleration of particle at $t = 2$ second is

(1) 2 m/s^2 (2) $\sqrt{20}$ m/s^2
(3) 4 m/s^2 (4) 5 m/s^2

26. A particle starts from the origin at $t = 0$ with a velocity $\left(10\hat{i}\right)$ m s^{-1} and moves in x-y plane with acceleration of $\left(2\hat{i}+2\hat{j}\right)$ m s^{-2}. Then at $t = 2$ s, match the enteries in column I with column II (in SI unit).

	Column I		Column II
a.	x-coordinate is	(i)	$2\sqrt{2}$
b.	y-coordinate is	(ii)	$2\sqrt{53}$
c.	Speed of the particle is	(iii)	24
d.	Magnitude of acceleration of the particle is	(iv)	4

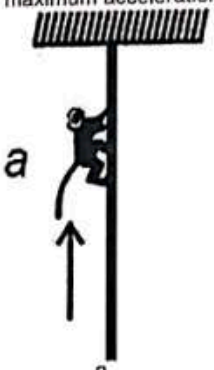
(1) a(i), b(ii), c(iii), d(iv) (2) a(iii), b(iv), c(ii), d(i)
(3) a(ii), b(iv), c(i), d(iii) (4) a(iv), b(iii), c(ii), d(i)

27. A projectile can have same range from two angles of projection with same initial speed. If 16 m and 9 m be the maximum heights, then range R is given by

(1) 20 m (2) 10 m
(3) 24 m (4) 48 m

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28. A monkey of mass 30 kg climbs up on a light rope as shown in figure. If breaking strength of the rope is 400 N, then maximum acceleration of the monkey will be



(1) 5 m/s^2 (2) 20 m/s^2
(3) Zero (4) $\frac{10}{3}$ m/s^2

29. A machine gun fires a bullet of mass 50 g with velocity 500 m/s. If average force of 200 N is acting on the gun, then maximum number of bullet fired per second will be

(1) 6 (2) 5
(3) 8 (4) 10

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30. Two springs A and B having force constant K_A and K_B ($K_A = 3K_B$), are stretched by applying equal forces. If energy stored in A is E, then that in B is

(1) $\frac{E}{3}$ (2) E
(3) $3E$ (4) $9E$

31. Two particles, A and B of equal masses having initial velocities $\overrightarrow{v_A} = \left(8\hat{i}+8\hat{j}\right)$ m/s and $\overrightarrow{v_B} = \left(\hat{i}+\hat{j}\right)$ m/s respectively collide. If the collision is head-on and elastic, then the final velocities of A and B respectively will be

(1) $\left(3\hat{i}+4\hat{j}\right)$ m/s and $\left(\hat{i}+\hat{j}\right)$ m/s
(2) $\left(\hat{i}+\hat{j}\right)$ m/s and $\left(7\hat{i}+7\hat{j}\right)$ m/s
(3) $\left(\hat{i}+2\hat{j}\right)$ m/s and $\left(2\hat{i}+\hat{j}\right)$ m/s
(4) $\left(\hat{i}+\hat{j}\right)$ m/s and $\left(8\hat{i}+8\hat{j}\right)$ m/s

32. The velocity with which an object should be projected from the surface of earth, so that its maximum height above the surface becomes equal to half the radius of earth will be (where R denotes radius of earth and g denotes acceleration due to gravity on surface of earth)

(1) $\sqrt{\frac{2gR}{3}}$
(2) $\sqrt{2gR}$
(3) $\sqrt{\frac{3gR}{2}}$
(4) $\sqrt{\frac{2gR}{5}}$

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33. An artificial satellite is moving in a circular orbit of radius r around a planet. Total energy of satellite is E. Energy required to move this satellite into a new orbit of radius $2r$, is
(1) $\frac{E}{4}$ (2) $\frac{-E}{4}$
(3) $\frac{E}{2}$ (4) $\frac{-E}{2}$

34. Moment of inertia of ring about a tangential axis perpendicular to the plane of ring will be (Assume the mass of ring to be M and radius to be R)
(1) MR^2 (2) $\frac{MR^2}{2}$
(3) $\frac{3MR^2}{2}$ (4) $2MR^2$

35. Keeping the mass of earth to be constant if its radius changes from R to $\frac{R}{\sqrt{3}}$, then new duration of the day will be
(1) 6 hours (2) 8 hours
(3) $8\sqrt{3}$ hours (4) $24\sqrt{3}$ hours

36. A particle of mass 300 g is moving with a speed of 20 ms^{-1} along the straight line $y = x - 4\sqrt{2}$. The angular momentum of the particle about the origin is (where y & x are in metres)
(1) 24 kg m^2s^{-1} (2) $24\sqrt{2}$ kg m^2s^{-1}
(3) 12 kg m^2s^{-1} (4) $6\sqrt{2}$ kg m^2s^{-1}

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37. Consider the following statements
Statement (A): Net positive work on a body increases its kinetic energy.
Statement (B): Net negative work on a body decreases its kinetic energy.
Based on above information pick the correct option.
(1) Both statements (A) and (B) are correct
(2) Both statement (A) and (B) are incorrect
(3) Statement (A) is correct while statement (B) is incorrect
(4) Statement (A) is incorrect while statement (B) is correct

38. Two different objects of masses m_1 and m_2 are allowed to fall from rest and from same point O along two different frictionless paths. The speed of the objects on reaching the ground will be in ratio
(1) $m_1 : m_2$ (2) $m_1 : 2m_2$
(3) 1 : 1 (4) $m_1 : 3m_2$

39. A particle starts from origin at $t = 0$ with velocity $\left(4\,\hat{i}\ m/s\right)$ and moves in x-y plane with constant acceleration of $\left(6\hat{i} + 8\hat{j}\right)\ m/s^2$. The x-coordinate of particle in 3 s will be equal to
(1) 13 m (2) 17 m
(3) 34 m (4) 39 m

40. A car travelling with a speed of 40 km/h can stop within a distance of 30 m when brakes are applied. If the car is going twice as fast i.e., 80 km/h, the stopping distance will be
(1) 60 m (2) 30 m
(3) 80 m (4) 120 m

41. The most precise reading of the length of an object, among the following is
(1) 10 cm (2) 10.0 cm
(3) 10.01 cm (4) 10.001 cm

42. Position of particle A moving along x-axis is given as $x = 4t^2$ and particle B moving along y-axis is given as $y = 3t^2$, where x and y are in metre and t is in second. The speed of particle A w.r.t. the particle B at $t = 1$ s will be
(1) 6 m/s (2) 4 m/s
(3) 10 m/s (4) 8 m/s

43. A uniform ladder of length 5 m is placed against the wall as shown in the figure. If coefficient of friction between ladder and floor is μ while wall is smooth then the minimum value of μ for ladder to not slip, will be



(1) $\mu = \frac{1}{8}$ (2) $\mu = \frac{3}{4}$
(3) $\mu = \frac{1}{3}$ (4) $\mu = \frac{3}{5}$

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44. The escape velocity from the Earth's surface is v. The escape velocity from the surface of another planet having a radius, four times that of Earth and same mass density is
(1) $4v$ (2) v
(3) $2v$ (4) $3v$

45. Position of a body moving along x-axis varies with time as $x = 12t - t^3 + 8$. The acceleration of the body when its velocity becomes zero is
(1) 6 m/s^2 (2) -12 m/s^2
(3) 8 m/s^2 (4) -4 m/s^2

CHEMISTRY

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46. Choose the incorrect match:

Base Physical Quantity	SI unit
(1) Electric current	Ampere
(2) Amount of substance	Kilogram
(3) Thermodynamic temperature	Kelvin
(4) Length	Metre

(1) (1) (2) (2)
(3) (3) (4) (4)

47. The number of significant figures in 2.005 is
(1) 1 (2) 2
(3) 3 (4) 4

48. Law of Multiple proportion is applicable to which pair of compounds?
(1) H_2S and H_2O (2) HCl and HBr
(3) SO_2 and SO_3 (4) NaCl and KCl

49. An element X has the following isotopic composition ^{60}X : 60% and ^{62}X : 40%. The weighted average atomic mass of the naturally occurring element X is
(1) 60.4 amu (2) 60.5 amu
(3) 60.2 amu (4) 60.8 amu

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50. Match List I with List II and choose the correct option.

	List I (Amount of Substance)		List II (Number of oxygen atoms)
a.	18 g $C_6H_{12}O_6$	(i)	4 N_A
b.	44.8 L NO_2(g) at STP	(ii)	0.03 N_A
c.	1.225 g $KClO_3$	(iii)	0.5 N_A
d.	11.2 L CO(g) at STP	(iv)	0.6 N_A

(1) a(iv), b(i), c(ii), d(iii) (2) a(i), b(ii), c(iii), d(iv)
(3) a(iv), b(iii), c(ii), d(i) (4) a(ii), b(iii), c(i), d(iv)

51. Consider the following statements
(a) 1 amu = 1.66056×10^{-27} kg
(b) 1 mole of atomic H contains 6.022×10^{-23} hydrogen atoms
(c) SI unit of luminous intensity is candela
The correct statements are
(1) (a) and (b) only (2) (b) and (c) only
(3) (a) and (c) only (4) (a), (b) and (c)

52. A compound contains 8% sulphur by mass, the minimum molecular mass of the compound is
(1) 800 u (2) 1200 u
(3) 200 u (4) 400 u

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53. An organic compound contains C, H and O only. If the percentage of C and H in it is 26% and 4.35% respectively, then the empirical formula of the compound is
(1) C_2H_4O (2) CH_2O_2
(3) CH_4O_2 (4) CH_2O

54. Consider the given reaction
$A + 2B \rightarrow 3C$
2 moles of each A and B have been taken in a closed container to form C.
Consider the following observations.
(a) A is limiting reagent
(b) 3 moles of C is formed
(c) 1 mole of B is left after the reaction.
The incorrect statement(s) is/are
(1) (b) only (2) (a) and (c) only
(3) (a) and (b) only (4) (b) and (c) only

55. If 3.01×10^{23} molecules of sugar are present in 24.5 mol of water then mole fraction of sugar in solution is
(1) 0.08 (2) 0.04
(3) 0.2 (4) 0.02

56. Incorrect statement regarding cathode rays is
(1) The cathode rays start from cathode and move towards the anode
(2) In the absence of electric and magnetic field these rays travel in straight line
(3) The charge to mass ratio of the cathode rays depends upon the gas from which they originate
(4) Cathode rays consist of negatively charged particles

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57. Consider the following statements
(a) d_{z^2} and $d_{x^2-y^2}$ orbitals have electron density along the axes.
(b) The maximum number of 3p electrons in an atom having spin quantum number of $-\frac{1}{2}$ is three.
(c) Shape of orbital is determined by Azimuthal quantum number.
(d) K^+, Ca^{2+} and S^{2-} are isoelectronic species.
(e) The number of protons, neutrons and electrons in $^{23}_{11}Na$, respectively are 11, 12 and 11.
The correct statements are
(1) (a), (b) and (c) only (2) (a) and (e) only
(3) (c), (d) and (e) only (4) (a), (b), (c), (d) and (e)

58. Which of the following species are isoelectronic?
(a) CH_3^+
(b) H_3O^+
(c) NH_3
(d) H_2O
Choose the correct option.
(1) (a) and (c) only (2) (c) and (d) only
(3) (a) and (b) only (4) (b), (c) and (d) only

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59. A 100 watt bulb emits monochromatic light of wavelength 400 nm. The number of photons emitted per second by the bulb is
(1) 1×10^{15} (2) 2×10^{20}
(3) 2×10^{10} (4) 8×10^{20}

60. Which experimental phenomenon is explained by wave nature of electromagnetic radiation?
(1) Diffraction
(2) Black-body radiation
(3) Line spectra of hydrogen atom
(4) Photoelectric effect

61. Match List-I with List-II.

List-I			List-II	
	(Lower energy Level) n_1	(Higher energy Level) n_2		Series
(a)	1	2, 3...	(i)	Brackett
(b)	4	5, 6...	(ii)	Paschen
(c)	3	4, 5...	(iii)	Pfund
(d)	5	6, 7...	(iv)	Lyman

The **correct** match is
(1) (a)-(i), (b)-(iv), (c)-(ii), (d)-(iii)
(2) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)
(3) (a)-(i), (b)-(iv), (c)-(iii), (d)-(ii)
(4) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)

62. If a particle of mass 20 mg is moving with a velocity of 500 ms^{-1} then the de Broglie wavelength associated with the moving particle is $(h = 6.6 \times 10^{-31}\ Js)$

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(1) 3.3×10^{-35} m (2) 6.6×10^{-32} m
(3) 6.6×10^{-35} m (4) 3.3×10^{-36} m

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63. Match List-I with List-II and choose the correct code.

	List-I (Orbitals)		List-II (Number of radial nodes)
(a)	2p	(i)	1
(b)	3p	(ii)	3
(c)	3s	(iii)	0
(d)	4s	(iv)	2

(1) (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)
(2) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
(3) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
(4) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

64. Which of the following sets of quantum numbers is not possible?
(1) n = 4, ℓ = 3, m = +3, s = $-\frac{1}{2}$
(2) n = 4, ℓ = 2, m = +3, s = $+\frac{1}{2}$
(3) n = 4, ℓ = 2, m = –2, s = $+\frac{1}{2}$
(4) n = 4, ℓ = 3, m = –2, s = $+\frac{1}{2}$

65. Expanded octet is present in
(1) CH_4 (2) $AlCl_3$
(3) SF_6 (4) NH_3

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66. Which among the following has highest lattice enthalpy?
(1) LiCl (2) NaCl
(3) LiI (4) NaI

67. Select the **incorrect** statement about resonance
(1) The resonating structures are hypothetical
(2) The number of unpaired electrons in various resonating structures remain same
(3) Hybrid structure is least stable
(4) Resonance involves delocalization of π- electrons

68. Match List-I with List-II

	List I (Molecule)		List II (Shape)
a.	ClF_3	(i)	Bent T-shape
b.	XeF_4	(ii)	Square planar
c.	PCl_5	(iii)	Trigonal pyramidal
d.	NH_3	(iv)	Trigonal bipyramidal

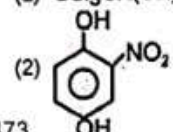
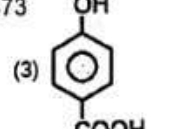
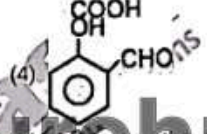
Choose the correct option.
(1) a(i), b(ii), c(iv), d(iii) (2) a(iii), b(iv), c(i), d(ii)
(3) a(i), b(iv), c(iii), d(ii) (4) a(iii), b(ii), c(iv), d(i)

69. Formal charge on the central oxygen atom in O_3 molecule is
(1) Zero (2) –1
(3) +1 (4) –2

70. The hybridisation of atomic orbitals of Xe in XeF_2, XeF_6 and XeO_3 respectively are
(1) sp^3d, sp^3d^2 and sp^3 (2) sp^2, sp^3d^2 and sp^3
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(3) sp^3d, sp^3d^3 and sp^3 (4) sp^2, sp^3d^3 and sp^3

71. Given below are two statements one is labelled as assertion (A) other is labelled as reason (R).
Assertion (A): O_2 is a paramagnetic species.
Reason (R): O_2 has two unpaired electrons in π-antibonding molecular orbital.
In the light of the above statements, choose the correct answer from the options given below.
(1) Both A and R are true and R is the correct explanation of A
(2) Both A and R are true but R is NOT the correct explanation of A
(3) A is true but R is false
(4) A is false but R is true

72. Species which does not contain σ bond is
(1) O_2 (2) C_2
(3) F_2 (4) N_2

73. Which of the following does not form intramolecular H-bonding?
(1) $CCl_3CH(OH)_2$
(2) 
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(3) 
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(4) 

74. Which among the following statement(s) is incorrect for physical equilibrium?
(1) Equilibrium is possible only in a closed system at a given temperature
(2) Both the opposing processes occur at the same rate and there is a static but stable condition
(3) All measurable properties of the system remain constant
(4) When equilibrium is attained for a physical process, it is characterised by constant value of one of its parameters at a given temperature

75. Match reaction in List-I with their correct relation of K_p and K_c in List-II

List-I	List-II
(a) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$	(i) $K_p = K_c(RT)^{-2}$
(b) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$	(ii) $K_p = K_c(RT)$
(c) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$	(iii) $K_p = K_c$
CC-473 (d) $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$	(iv) $K_p = K_c(RT)^{-1}$ CC-473

Choose the **correct** match
(1) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
(2) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)
(3) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)
(4) (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)

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76. Which of the following is an example of homogeneous equilibrium?

(1) $H_2O_{(l)} \rightleftharpoons H_2O_{(g)}$

(2) $CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$

(3) $H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$

(4) $2CO_{(g)} \rightleftharpoons CO_{2(g)} + C_{(s)}$

77. If K_C for the reaction, $N_2 + 3H_2 \rightleftharpoons 2NH_3$ is K_1, then K_C for the reaction, $NH_3 \rightleftharpoons \frac{1}{2}N_2 + \frac{3}{2}H_2$ is

(1) $\frac{1}{K_1}$ (2) $\frac{1}{\sqrt{K_1}}$

(3) K_1^2 (4) $\frac{1}{K_1^2}$

78. If equilibrium constant of a reaction at 27°C is 10 then the standard Gibbs free energy change for a reaction at equilibrium will be

(1) –3.54 kcal mol^{-1} (2) –2.12 kcal mol^{-1}

(3) –1.38 kcal mol^{-1} (4) Zero

79. $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

In order to produce high amount of PCl_3, the reaction should be subjected to

(1) Increase in pressure (2) Decreasing pressure

(3) Removal of PCl_5 (4) Addition of catalyst

80. The pH of 10^{-8} M NaOH solution is

(1) 6.98 (2) 6

(3) 7.02 (4) 8

81. Consider the following statements:

(i) According to Bronsted-Lowry concept of acid and base, H_2O is amphoteric in nature.

(ii) Conjugate acid and conjugate base of HCO_3^- is H_2CO_3 and CO_2 respectively.

(iii) HCl is conjugate acid of base Cl^-.

The correct statements among the following is/are

(1) (ii) only (2) (i) and (iii) only

(3) (ii) and (iii) only (4) (i), (ii) and (iii)

82. Which of the following salts undergoes cationic hydrolysis?

(1) NH_4Cl (2) $NaHCO_3$

(3) $NaNO_3$ (4) NaCN

83. When 50 mL of 0.1 M NH_4OH is mixed with 25 mL of 0.1 M HCl, then pH of the resulting mixture is [$pK_b = 4.74$]

(1) 4.74 (2) 9.26

(3) 7 (4) 10

84. The solubility (in mol/L) of Ag_2CrO_4 in a 0.2 M K_2CrO_4 solution is (Given : K_{sp} of Ag_2CrO_4 = 1.1×10^{-12} at 298 K and $\sqrt{5.5} = 2.34$)

(1) 1.17×10^{-5} (2) 1.17×10^{-6}

(3) 1.4×10^{-8} (4) 2.1×10^{-7}

85. Correct order of stability of +1 oxidation state of Tl, In, Ga and Al follows the order

(1) Al > Ga > In > Tl (2) In > Tl > Ga > Al

(3) Tl > In > Ga > Al (4) In > Ga > Tl > Al

86. The number of B–O bonds in borax is

(1) 14 (2) 5

(3) 4 (4) 12

87. Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R** :

Assertion (A): Boron is unable to form BF_6^{3-} ion.

Reason (R): Due to non-availability of *d*-orbitals boron is unable to expand its octet.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(3) (A) is correct but (R) is incorrect

(4) (A) is incorrect but (R) is correct

88. Which of the following statements is incorrect about diborane (B_2H_6)?

(1) In B_2H_6, four terminal hydrogen atoms and the two boron atoms lie in one plane

(2) In B_2H_6, two bridging hydrogen atoms are present

(3) In B_2H_6, there are two $2c - 2e^-$ bonds and four $3c - 2e^-$ bonds

(4) In B_2H_6, each boron atom is sp^3 hybridised

89. The element which do not show catenation property is

(1) C (2) Si

(3) Sn (4) Pb

90. In Buckminsterfullerene, the hybridisation of carbon is

(1) *sp* (2) sp^2

(3) sp^3 (4) dsp^2

BOTANY

91. Given below are the stages in the life cycle of a Fern plant.

(A) Development of young embryo takes place within the female gametophyte

(B) Fusion of gametes result in the formation of zygote

(C) Formation of antheridia and archegonia

(D) Development of inconspicuous, small, multi-cellular, free living, photosynthetic thalloid gametophyte

(E) Spore mother cell undergo sporic meiosis to produce spores

Choose the **correct** sequence of occurrence of these events from the option given below.

(1) A, B, D, E, C (2) E, D, C, B, A

(3) B, C, D, E, A (4) A, B, C, E, D

92. How many of the given statement is/are **true** w.r.t. space enclosed by inner membrane of chloroplast having enzymes for the synthesis of carbohydrate?

(a) It contains small, double stranded circular DNA and light harvesting complexes.

(b) In this region, ATP synthesis takes place through substrate level phosphorylation

(c) Presence of starch granules

(d) Presence of 30S smaller subunit in ribosome

(e) If light becomes unavailable, the process of carbohydrate formation stops here immediately

(1) One (2) Two

(3) Four (4) Five

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93. The layer of bacterial cell envelope which is structurally similar to that of eukaryotes is related to all of the given features, **except**
(1) Lipid bilayer mainly consists of phosphoglycerides
(2) Hydrophilic head is outside and hydrophobic tails are inside
(3) Tail is composed of only unsaturated hydrocarbons
(4) Within the bilayer, proteins show lateral movement

94. A membraneless cell organelle which is **not** found in higher plants is/has
(1) 9 + 2 arrangement of microtubules
(2) A role in formation of flagella in prokaryotes
(3) Composed of rRNA and proteins
(4) Involved in the formation of spindle fibres during cell division in animals

95. Which of the following statements is **incorrect** w.r.t. Endoplasmic Reticulum.
(1) RER is actively involved in protein synthesis and secretion
(2) SER is the major site for synthesis of lipids
(3) It divides the intracellular space into three distinct compartments
(4) In animal cells lipid like steroidal hormones are synthesised in SER

96. '*Omnis cellula-e cellula*' explains that
(1) Organisms are made up of cells and its products
(2) Presence of cell wall is a unique character of plant cells
(3) Activities of an organism are sum total of interactions of its constituent cells
(4) Cells arise from pre-existing cells

97. Chromosome in which centromere is present at the centre
(1) Is called acrocentric chromosome
(2) Does not have a primary constriction
(3) Appears L-shaped during anaphase
(4) Contains two equal arms

98. Match the columns and select the **correct** option.

	Column I		Column II
i.	Aleuroplast	(a)	Rich in carotenoids
ii.	Chromoplast	(b)	Stores fats
iii.	Chloroplast	(c)	Synthesis of carbohydrates
iv.	Elaioplast	(d)	Stores proteins

	i	ii	iii	iv
(1)	(c)	(a)	(d)	(b)
(2)	(b)	(d)	(a)	(c)
(3)	(b)	(c)	(d)	(a)
(4)	(d)	(a)	(c)	(b)

(1) (1) (2) (2)
(3) (3) (4) (4)

99. Kinetochores are
(1) Terminal ends of the chromosomes
(2) The structures that hold two chromatids of a chromosome
(3) The parts of chromosomes beyond the secondary constriction
(4) The disc shaped structures on the side of centromere

100. The middle lamella present between neighbouring cells of plants is mainly made up of
(1) Calcium carbonate (2) Calcium pectate
(3) Hemicellulose (4) Proteins

101. Thick and tough glycocalyx found in some bacteria is called
(1) Slime layer (2) Mesosome
(3) Capsule (4) Lamellae

102. How many doublets of microtubules are arranged peripherally in the axoneme of flagella?
(1) 18 (2) 22
(3) 20 (4) 9

103. Cytotaxonomy is based on all, **except**
(1) Chromosome number
(2) DNA sequence on chromosome
(3) Structure and behaviour of chromosome
(4) Cytological information

104. Which one is **wrong** statement w.r.t. bacteria?
(1) They are the sole members of kingdom Monera
(2) They are very complex in behaviour
(3) They show most extensive metabolic diversity
(4) They can be only parasite or saprophyte

105. Archaebacteria differ from other bacteria as the former have
(1) Different cell wall structure
(2) Membrane bound organelles
(3) 70S ribosomes
(4) Genetic material not enclosed in nuclear membrane

106. According to the five kingdom classification, chrysophytes, dinoflagellates, euglenoids, slime moulds and protozoans fall under the kingdom
(1) Whose members are primarily aquatic and some have flagella and cilia
(2) In which the members are believed to be the primitive relatives of prokaryotic life forms.
(3) Whose members reproduce asexually by a process involving zygote formation.
(4) Whose boundaries are well defined and it does not form any link with plants, animals and fungi

107. Read the following statements and state them as true **(T)** or false **(F)** and select the **correct** option.
a. Diatoms float passively in water currents.
b. Reserve food material is stored in the form of oil and glycogen in fungi.
c. Cyanobacteria are characterised by the presence of flagellum throughout the life cycle.
d. Kingdom fungi is classified only on the basis of fruiting bodies into various classes.
e. Deuteromycetes are commonly known as imperfect fungi.
(1) a-(T), b-(T), c-(F), d-(F), e-(T)
(2) a-(F), b-(F), c-(T), d-(T), e-(F)
(3) a-(T), b-(F), c-(T), d-(F), e-(F)
(4) a-(F), b-(T), c-(T), d-(F), e-(T)

108. Read the following statements and select the option with all the **correct** statements.
(A) Sexual reproduction in Monera involves cell fusion and gamete formation.
(B) Disease caused by *Plasmodium* has a staggering effect on the human population.
(C) *Saccharomyces* being Ascomycete have septate and branched mycelium.
(D) Coenocytic hyphae are continuous tubes filled with multinucleated cytoplasm.
(1) (A) and (B) (2) (B) and (C)
(3) (C) and (D) (4) (B) and (D)

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109. Read the following features of a cell organelle.
a. Number of this organelle per cell is variable depending on physiological activity of the cell
b. Requires specific vital stain to be viewed under the microscope
c. Their palade particles are same as prokaryotes
d. Sausage shaped or cylindrical
Identify the cell organelle on the basis of above features.
(1) Chloroplast (2) Nucleus
(3) Mitochondria (4) ER

110. How many of the following statements is/are **correct** for chromatin material of eukaryotes?
(a) Composed of DNA and histone proteins.
(b) Euchromatin is the loosely packed and transcriptionally active region.
(c) At interphase, chromatin material gets condensed into chromosomes.
(d) It can be stained by basic dyes.
(1) One (2) Two
(3) Three (4) Four

111. Read the following statements Assertion (A) and Reason (R) and choose the **correct** option.
Assertion (A) : The spread of living pteridophytes is limited and restricted to narrow geographical regions.
Reason (R) : The gametophytes of pteridophytes require cool, damp, shady places to grow and need water for fertilization.
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

112. Select the **wrongly** matched pair.
(1) *Ectocarpus* – Laterally attached flagella in zoospores
(2) *Sphagnum* – Zygotic meiosis absent
(3) *Funaria* – Vegetative reproduction by fragmentation
(4) *Porphyra* – Motile asexual spores

113. How many of the given statement(s) is/are **incorrect**?
a. At least half of the total carbon dioxide fixation on earth is carried out by algae through photosynthesis.
b. Mosses generally are of little ecological importance.
c. *Chlorella* is used as food supplement by space travellers.
d. Plant body of bryophytes is more differentiated than that of algae.
(1) One (2) Two
(3) Three (4) Four

114. How many of the following statements are correct?
a. Pollen grain is reduced male gametophyte in gymnosperms.
b. Archegonia is female sporophyte in pteridophytes.
c. The male or female strobili are borne on the same tree in *Cycas*.
d. Ovules are present only in angiosperms.
e. In gymnosperms, pollens are transferred via air currents.
(1) Three (2) Two
(3) Five (4) Four

115. The food is stored as floridean starch in
(1) *Ectocarpus* (2) *Polysiphonia*
(3) *Chara* (4) *Spirullina*

116. The non vascular terrestrial plants
(1) Are first embryophytes
(2) Possess diploid spores
(3) Are heterosporous
(4) Have free living sporophyte

117. Select the feature which is **not true** for red wood tree.
(1) Seeds are naked
(2) One of the tallest angiosperm
(3) Have archegonia inside ovule
(4) Heterosporous

118. Which of the given statements are **true** regarding PS I and PS II?
a. Both are found on stroma lamellae.
b. Both are involved in non-cyclic flow of electrons.
c. Only PS II is associated with release of O_2.
d. PS II lies on outer surface whereas PS I on inner surface of thylakoids.
(1) Only b & c (2) All a, b, c & d
(3) Only b, c & d (4) Only c & d

119. How many ATP are required to form a molecule of hexose sugar through photosynthesis in C_4 plants?
(1) 18 (2) 12
(3) 30 (4) 45

120. Match the following columns and select the **correct** option.

	Column-I		Column-II
a.	Chemiosmotic hypothesis	(i)	Melvin Calvin
b.	First action spectrum of photosynthesis	(ii)	Joseph priestley
c.	C_3 cycle	(iii)	P. Mitchell
d.	Discovered oxygen	(iv)	T.W. Engelmann

(1) a(iii), b(i), c(iv), d(ii) (2) a(ii), b(i), c(iv) d(iii)
(3) a(iii) b(iv), c(i), d(ii) (4) a(i), b(iii), c(iv), d(ii)

121. Match column-I with column-II and select the **correct** option.

Column-I		Column-II
a. Phosphoglycolate	(i)	Undergoes decarboxylation in bundle sheath cells
b. 3-phosphoglycerate	(ii)	Formed when the most abundant enzyme in the world binds with oxygen
c. Aspartic acid	(iii)	Regenerates in the Mesophyll of C_3 plant by using ATP
d. RuBP	(iv)	Product of the most crucial step of the Calvin cycle

Choose the **correct** option:
(1) a-(i), b-(iv), c-(iii), d-(ii) (2) a-(ii), b-(i), c-(iv), d-(iii)
(3) a-(iii), b-(ii), c-(i), d-(iv) (4) a-(ii), b-(iv), c-(i), d-(iii)

122. Tomato and maize can be differentiated from each other as
(a) For tomato, the current availability of CO_2 levels is limiting.
(b) The latter responds to high CO_2 concentrations under low light conditions.
(c) In the former, water stress leads to reduction in the CO_2 availability.
(d) In the latter, RuBisCO shows only carboxylase activity.
(e) In the former, PEPcase is found in mesophyll cells for the carboxylation of RuBP.
The **correct** ones are:
(1) (b) and (e) only (2) (a) and (d) only
(3) (a), (c) and (d) only (4) (c) and (e) only

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123. Mark the **correct** statements w.r.t. the pigment 'X' that appears blue green in the chromatogram.
(a) 'X' is the chief pigment associated with photosynthesis.
(b) In PS-I the reaction center 'X' has an absorption peak at 680 nm.
(c) 'X' is also a shield pigment.
(d) 'X' shows maximum absorption in the blue region of the spectrum only.
(e) 'X' helps to make photosynthesis more efficient by absorbing all different wavelengths of light.
(1) (a), (d), (e) are correct (2) (a) and (b) are correct
(3) (d) and (e) are correct (4) Only (a) is correct

124. Kranz anatomy is found in the leaves of
(1) Tomato (2) Sugarcane
(3) Rice (4) Wheat

125. In plants sucrose can be broken down into glucose and fructose with the help of the enzyme
(1) Zymase (2) Cellulase
(3) Invertase (4) Pectinase

126. Match the following complexes to their alternate name and select the **correct** option.

Column I		Column II
a. Complex I	(i)	Cytochrome bc_1 complex
b. Complex II	(ii)	Cytochrome c oxidase complex
c. Complex III	(iii)	NADH dehydrogenase
d. Complex IV	(iv)	Succinate dehydrogenase

(1) a(iii), b(iv), c(i), d(ii) (2) a(ii), b(iv), c(i), d(iii)
(3) a(ii), b(iii), c(iv), d(i) (4) a(iii), b(i), c(ii), d(iv)

127. During which of the following conversions, the first decarboxylation in Krebs cycle occurs?
(1) Oxaloacetic acid to citric acid
(2) α-ketoglutaric acid to succinyl CoA
(3) Isocitric acid to oxalosuccinic acid
(4) Oxalosuccinic acid to α-ketoglutaric acid

128. Choose the **correct** set of mineral element and enzymes required for the given below equation.

$Pyruvic\ acid + CoA + NAD^+ \rightarrow Acetyl\ CoA + CO_2 + NADH + H^+$

(1) Mn^{2+}, pyruvate decarboxylase
(2) Mg^{2+}, pyruvate dehydrogenase
(3) Mn^{2+}, pyruvate dehydrogenase
(4) Cu^{2+}, pyruvate decarboxylase

129. How many ATP will be produced by complete oxidation of one molecule of acetyl CoA?
(1) 6 (2) 8
(3) 2 (4) 12

130. State **true (T)** or **false (F)** for the following statements.
a. Yeasts poison themselves to death when the concentration of alcohol reaches about 10%.
b. In fermentation, less than seven percent of the energy in glucose is released.
c. $NADH + H^+$ is reoxidised to NAD^+ in lactic acid and alcoholic fermentation.
d. Pyruvic acid is reduced to lactic acid by homofermentative lactic acid bacteria.
(1) a(T), b(T), c(F), d(F) (2) a(F), b(T), c(F), d(F)
(3) a(F), b(F), c(T), d(F) (4) a(F), b(T), c(T), d(T)

131. Which one is a 5 C intermediate of TCA cycle?
(1) Citric acid (2) Succinyl CoA
(3) α-Ketoglutaric acid (4) Oxaloacetic acid

132. The final hydrogen acceptor in electron transport system in mitochondria is
(1) Cytochrome *a* (2) ADP
(3) Cytochrome *c* (4) Oxygen

133. All of the following easily multiply by fragmentation, **except**
(1) Fungi (2) Filamentous algae
(3) Protonema of moss (4) Unicellular algae

134. Match column-I with column-II and select the correct option.

Column-I		Column-II
a. *Albugo*	(i)	Fungi extensively used in biochemical and genetic work
b. *Neurospora*	(ii)	Fungi causing smut
c. *Ustilago*	(iii)	The parasitic fungi on mustard
d. *Alternaria*	(iv)	Fungi which show only vegetative phase

(1) a(i), b(iv), c(iii), d(ii) (2) a(iii), b(i), c(ii), d(iv)
(3) a(ii), b(iii), c(iv), d(i) (4) a(iii), b(ii), c(i), d(iv)

135. Read the following statements and select the **correct** option.
Statement 1 : The plasmid DNA confers certain unique phenotypic characters in bacteria.
Statement 2 : The plasmid DNA is used to monitor bacterial transformation with foreign DNA.
(1) Only statement 1 is correct
(2) Only statement 2 is correct
(3) Both statements 1 and 2 are correct
(4) Both statements 1 and 2 are incorrect

ZOOLOGY

136. CO_2 and H_2O reacts with each other in the presence of carbonic anhydrase to produce
(1) 6×10^5 molecules of H_2CO_3 per minute
(2) 6×10^5 molecules of H_2CO_3 per second
(3) 60×10^5 molecules of H_2CO_3 per minute
(4) 60×10^5 molecules of H_2CO_3 per second

137. A common feature of peptide and glycosidic bond is that they
(1) Connect carbohydrate moieties
(2) Are formed by dehydration reaction
(3) Are present in chitin
(4) Connect amino acids

138. Consider the statements given below:
Statement A: The proteins of every organism are considered as heteropolymers.
Statement B: Disulfide bonds play a special role in the structures of many proteins by forming covalent links between parts of a polypeptide molecule or between two different polypeptide chains.
Select the **correct** option:
(1) Only statement A is correct
(2) Only statement B is correct
(3) Both statements A and B are correct
(4) Both statements A and B are incorrect

139. The components of a Lecithin molecule are all of the following, **except**
(1) Nitrogen containing base (2) Glycerol
(3) Fatty acids (4) Cholesterol

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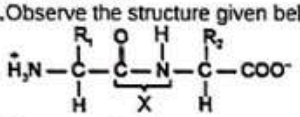
140. Select the molecule that is formed by dehydration process and contains two heterocyclic rings in its structure.
(1) Uridine (2) Guanine
(3) Cholesterol (4) Cytosine

141. A linear polysaccharide is composed of 458 glucose monomers connected exclusively by glycosidic linkages. During its complete enzymatic hydrolysis, which of the following options accurately reflects the above given reaction?
(1) Upon complete hydrolysis, 458 glycosidic bonds will be broken
(2) A total of 457 water molecules will be consumed for the complete hydrolysis of glycosidic bonds.
(3) Only 457 glucose molecules will be liberated as the final product.
(4) A total of 458 water molecules will be released after hydrolysis of glycosidic bonds.

142. Which of the following carbohydrates can be broken into simpler units *via* the action of the enzymes that have the EC number 3.2.1.20?
(1) Ribose (2) Glucose
(3) Galactose (4) Maltose

143. If we plot a graph for analysing the variation of change in temperature (x-axis) with enzyme activity (y-axis), the curve obtained will be CC-473
(1) Bell-shaped (2) Sigmoid
(3) Parabolic (4) Linear

144. Observe the structure given below.



X represents a/an
(1) Ionic linkage (2) Peptide bond
(3) Glycosidic bond (4) Ester bond

145. **Assertion (A):** Inhibition of succinic dehydrogenase by malonate is an example of competitive inhibition.
Reason (R): Malonate closely resembles the substrate of succinic dehydrogenase in the structure.
In the light of the above statements, choose the correct option.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) (A) is false but (R) is true

146. How many statements are **incorrect** w.r.t. the enzymes?
(a) Enzymes are divided into 6 classes, each with 4-13 subclasses and named accordingly by a 4-digit number.
(b) Ligases catalyse the joining of C – O, C – S, C – N, P – O etc bonds.
(c) Hydrolases catalyse the hydrolysis of ester, ether, peptide and glycosidic bonds.
(d) Catalytic activity is lost permanently when the enzymes are exposed to low temperature.
(1) Two (2) Three
(3) Four (4) One

147. Select the option that fills the blank **correctly**.
The sequence in which amino acids are arranged or their positional information in a protein *i.e.* which one is the first amino acid, which one is second and so on is given by the _____ structure of a protein.
(1) Primary (2) Secondary
(3) Tertiary (4) Quaternary

148.

Glucose, Cytidylic acid, Alanine, Lecithin, Ribose, Starch, Myoglobin, Glycine, Guanylic acid

How many of them are micromolecules and are present in filtrate during analysis of biomolecules?
(1) Five (2) Six
(3) Seven (4) Nine

149. Read the given statements and state them as true (T) or false (F).
a. Antibody does not fight against infectious agents.
b. GLUT-4 enables glucose transport into cells.
c. Trypsin is an active proteolytic enzyme.
Choose the suitable option.

	a	b	c
(1)	F	F	F
(2)	T	F	F
(3)	F	T	T
(4)	T	T	T

(1) (1) (2) (2)
(3) (3) (4) (4)

150. Match column I with column II and choose the correct option.

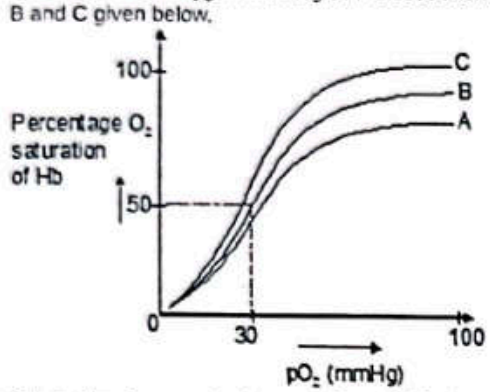
	Column I		Column II
(a)	Arachidonic acid	(i)	Trihydroxypropane
(b)	Palmitic acid	(ii)	Unsaturated fatty acid
(c)	Glycerol	(iii)	Saturated fatty acid

(a) (b) (c)
(1) (i) (iii) (ii)
(2) (iii) (i) (ii)
(3) (ii) (iii) (i)
(4) (ii) (i) (iii)
(1) (1) (2) (2)
(3) (3) (4) (4)

151. Diffusion membrane for gaseous exchange at alveolar surface lacks
(1) Basement substance between alveolar and capillary walls
(2) Simple squamous epithelium of alveoli
(3) Endothelium of alveolar capillaries
(4) Ciliated columnar epithelium

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152. There are three oxygen haemoglobin dissociation curves A, B and C given below.



Select the **incorrect** statement from following options if curve B represents oxygen haemoglobin dissociation curve when individual is at rest

(1) In case of curve C, the pO_2 at which haemoglobin is 50% saturated with O_2 is less than 30 mmHg

(2) Increase in temperature shifts curve 'B' towards right as represented by curve A which reflects increase in P_{50} value

(3) Curve C represents a possible increase in pH which indicates increase in affinity of haemoglobin for oxygen

(4) Increase in P_{50} value of haemoglobin as observed in curve A indicates enhanced association of O_2 with hemoglobin to form oxyhaemoglobin

153. Read the following statements carefully and select the **incorrect** option.

(1) CO_2 combines with haemoglobin at tissue level to form carbamino-hemoglobin.

(2) CO_2 is removed from blood at the alveolar level where pO_2 is low and pCO_2 is high in lungs.

(3) Zinc is a co-factor of enzyme carbonic anhydrase.

(4) Every 100 mL of oxygenated blood delivers around 5 ml of O_2 to tissues at rest under normal condition.

154. Read following statements carefully and select the option containing all **correct** statements.

a. pO_2 of atmospheric air is more than pO_2 of alveolar air.

b. Diffusion of gases through respiration membrane depends upon their solubility and partial pressure.

c. Intrapleural pressure during normal inspiration is less than intra pulmonary pressure.

d. Decrease in activity / frequency of pneumotaxic centre increases breathing rate.

(1) a, b and c (2) a, b and d

(3) b, c and d (4) a, c and d

155. Match column A with column B and select the **correct** option.

	Column A		Column B
a.	Medulla oblongata	(i)	Chemoreceptors
b.	Pons varolii	(ii)	'Switch-off' centre for breathing
c.	Aortic arch and carotid arch	(iii)	Normal respiratory rhythm centre

(1) a(ii), b(iii), c(i) (2) a(ii), b(i), c(iii)

(3) a(iii), b(ii), c(i) (4) a(iii), b(i), c(ii)

156. Which of the following factors favour decrease in breathing rate?

(1) Increased blood oxygen

(2) Increase blood CO_2

(3) Increased metabolism

(4) Increase in acidity of body fluid

157. Match the items given in column I to the items given in column II and select the **correct** option.

Column I	Column II
a. FRC	(i) ERV + RV
b. VC	(ii) RV + ERV + IRV + TV
c IC	(iii) TV + IRV
d. TLC	(iv) ERV + TV + IRV

(1) a(i), b(iv), c(iii), d(ii) (2) a(i), b(ii), c(iii), d(iv)

(3) a(i), b(iv), c(ii), d(iii) (4) a(i), b(ii), c(iv), d(iii)

158. CO_2 is transported from tissues to alveoli through blood in the plasma as well as RBCs. The major mode of transport of CO_2 through plasma is

(1) Sodium bicarbonate form (2) Carbamino-hemoglobin

(3) Carbonic acid form (4) Carboxyhemoglobin

159. Arrange the following given steps sequentially w.r.t. the process of respiration.

a. Transport of gases by the blood.

b. Diffusion of O_2 and CO_2 between blood and tissues.

c. Diffusion of gases (O_2 and CO_2) across alveolar membrane.

d. Pulmonary ventilation by which atmospheric air is drawn in and CO_2 rich alveolar air is released out.

e. Utilization of O_2 by the cells for catabolic reactions and resultant release of CO_2.

Choose the correct option.

(1) a → b → c → d → e (2) d → c → a → b → e

(3) c → a → e → d → b (4) d → a → b → c → e

160. During normal quiet inspiration

(1) External intercostal muscles relax

(2) Internal intercostal muscles contract

(3) Intra-pulmonary pressure is more than atmospheric pressure

(4) Volume of thoracic chamber increases in both antero-posterior and dorso-ventral axis

161. A loose connective tissue is composed of all of the following, **except**

(1) Matrix, cells and fibres

(2) Mixture of modified polysaccharides

(3) Different types of cells present in matrix

(4) Only one type of fibres

162. A histologist observes an epithelium containing single layer of tall and slender cells, having basal nuclei with mucous secreting cell in between them, on a slide through microscope.

Epithelium being discussed in the above the statement is

(1) Simple squamous epithelium

(2) Simple cuboidal epithelium

(3) Stratified squamous epithelium

(4) Columnar glandular epithelium

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163. **Assertion(A):** The action of muscle tissue moves the body to adjust to the changes in the environment and to maintain the positions of the various parts of the body.
Reason(R): The muscle fibres lengthen in response to stimulation, then shorten and return to their uncontracted state in a coordinated fashion.
In the light of above statements, choose the **correct** option.
(1) Both (A) and (R) are true and (R) explains (A) correctly
(2) Both (A) and (R) are true but (R) does not explain (A) correctly
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

164. Choose the **correct** match w.r.t. *Periplaneta*.
(1) Metathoracic wings – Opaque, dark and leathery
(2) Mesothoracic wings – Arise from 2nd thoracic segment
(3) Hindwings – Also called tegmina
(4) Forewings – Transparent and membranous

165. Select the **incorrect** statement w.r.t. cockroach.
(1) Gaseous exchange takes place at tracheoles by diffusion.
(2) Crop is the largest part of foregut where food is stored.
(3) Vision of cockroach is mosaic vision.
(4) A pair of ovaries lie laterally in 8th to 12th abdominal segments of female cockroach.

166. During hibernation (A) and aestivation (B), gaseous exchange in frog takes place respectively through

	A	B
(1)	Cutaneous respiration	Pulmonary respiration
(2)	Pulmonary respiration	Cutaneous respiration
(3)	Buccal respiration	Buccal respiration
(4)	Cutaneous respiration	Cutaneous respiration

(1) (1) (2) (2)
(3) (3) (4) (4)

167. Choose the **incorrect** statement.
(1) Testes in male frogs release sperms into the Bidder's canal, which then pass into the urinary bladder through ureters before passing out as cloacal discharge
(2) In frogs, both ureters and urinary bladder are present
(3) In female frogs, ovaries are situated near kidneys but lack any direct functional association with them
(4) In adult frogs, urea as chief excretory nitrogenous waste are carried into the kidney, where it is separated and excreted

168. In *Rana tigrina*, 'A' is responsible for transporting deoxygenated blood from various parts of the body towards the heart. 'A' empties into a triangular, thin-walled collecting chamber named 'B', which in turn drains the blood into the right atrium.
Choose the option that **correctly** represents the anatomical structures 'A' and 'B', respectively.
(1) Vein; Conus arteriosus
(2) Artery; Sinus venosus
(3) Vena cava; Sinus venosus
(4) Aorta; Conus arteriosus

169. Metameric segmentation, open circulation with internal fertilisation is **not** found in
(1) *Anopheles* (2) *Hirudinaria*
(3) *Periplaneta* (4) *Ascaris*

170. Triploblastic with organ system grade, coelomate and unsegmented animals are seen in
(1) Echinodermata (2) Arthropoda
(3) Ctenophora (4) Platyhelminthes

171. Choose the **correct** statement.
(1) Aquatic annelids like *Pila* possess lateral appendages, parapodia, which help in swimming.
(2) The body of arthropods is covered by calcareous shell.
(3) In sponges, the body is supported by skeleton made up of spicules or spongin fibres.
(4) *Hydra* exhibits metagenesis.

172. Select the **correct** set of animals which belongs to same phylum
(1) *Sycon* and *Hydra* (2) *Obelia* and *Physalia*
(3) *Pleurobrachia* and *Culex* (4) *Asterias* and *Ctenoplana*

173. Match the following organisms with their respective characteristics.

(a)	*Taenia*	(i)	Comb plates
(b)	*Ctenoplana*	(ii)	Malpighian tubules
(c)	*Locusta*	(iii)	Visceral hump
(d)	*Octopus*	(iv)	Hooks and suckers

Choose the **correct** option from the following
(1) a(iv), b(i), c(ii), d(iii) (2) a(i), b(ii), c(iii), d(iv)
(3) a(ii), b(iii), c(iv), d(i) (4) a(iii), b(iv), c(i), d(ii)

174. A college student was studying the distribution of germinal layers in an animal and found out that its body cavity is not lined by mesoderm instead it is present as scattered pouches.
Select the animal which has the feature observed by the student.
(1) *Pleurobrachia* (2) *Spongilla*
(3) *Ascaris* (4) *Nereis*

175. Match list I with list II and select the **correct** option.

	List I		List II
a.	*Myxine*	(i)	4 pairs of gills with gill cover
b.	*Carcharodon*	(ii)	Tetrapod with epidermal scales
c.	*Calotes*	(iii)	Absence of air bladder
d.	*Pterophyllum*	(iv)	Presence of more than four pairs of gill-slits

(1) a(iv), b(ii), c(i), d(iii) (2) a(iv), b(iii), c(ii), d(i)
(3) a(iv), b(ii), c(iii), d(i) (4) a(iii), b(iv), c(i), d(ii)

176. Adult lamprey differs from magur in having
(a) Paired fins
(b) Scales
(c) Marine habitat
(d) Parasitic mode of life
Select the **correct** option.
(1) (a) and (b) (2) (c) and (d)
(3) (a), (b) and (c) (4) (a), (b), (c) and (d)

177. Which of the following features is shared by both *Hemidactylus* and *Psittacula*?
(1) Completely divided four chambered heart
(2) Pneumatic bones
(3) Oviparous and show direct development
(4) Poikilotherms

178. As compared to *Clarias*, *Trygon* has
(1) Bony endoskeleton
(2) Electric organ
(3) Claspers as copulatory structure
(4) Presence of swim bladder

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179. How many of the organisms given in the box below are viviparous and show internal fertilisation with direct development?

Hyla, Hemidactylus, Testudo, Betta, Columba, Camelus, Canis, Ornithorhynchus

(1) Five
(2) Two
(3) Three
(4) Four

180. Choose the **incorrect** statement w.r.t. notochord.

(1) It is present throughout the life in lancelet
(2) It is present only in larval tail of *Doliolum*
(3) It is absent throughout the life cycle in shark
(4) It is replaced by a vertebral column in *Hyla*

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MM : 720 Project Phoenix for Re-NEET-2026_PST-01P **Time : 180 Min.**

PHYSICS

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CHEMISTRY

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BOTANY

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ZOOLOGY

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175. (2)
176. (2)
177. (3)
178. (3)
179. (2)

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157. (1)

158. (1)

180. (3)

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Hints and Solutions

PHYSICS

(1) **Answer :** (4)
Solution:
In 427.003 all digits are significant

(2) **Answer :** (1)
Solution:
In division the number of significant figure in final result is equal to minimum number of significant figures in dividend and divisor.
$\rho = \frac{42.60}{1.2}$ g cm^{-3} = 35.5 g cm^{-3}
Rounding off to 2 significant digits
ρ = 36 g cm^{-3}

(3) **Answer :** (2)

(4) **Answer :** (4)
Solution:
When a body is moving with constant speed, its tangential acceleration is zero because there is no change in the magnitude of velocity. However, the body can still have centripetal acceleration if it is moving in a circular path, which means the direction of velocity is changing. This centripetal acceleration can be of constant magnitude if the body is moving in a circle with constant radius and speed. Therefore, the acceleration of the body may be varying, may be zero, or may be of constant magnitude depending on the path of motion.

(5) **Answer :** (3)
Solution:
$t = \frac{d}{\sqrt{V_b^2 - V_R^2}} = \frac{200}{\sqrt{5^2 - 3^2}} = 50\ \text{s}$

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(6) **Answer :** (2)
Solution:
$v = \sqrt{2}v\cos\theta$
$\cos\theta = \frac{1}{\sqrt{2}}$
$\boxed{\theta = 45^\circ}$

(7) **Answer :** (1)
Solution:
Force = $\frac{\text{Change in momentum}}{\text{impact time}}$
$\left|\Delta \vec{p}_y\right| = 0 \quad \left|\Delta \vec{p}_x\right| = \left|-mv_2\sin\theta - mv_1\sin\theta\right|$
= 2 *mv* sinθ
Force = $\frac{\left|\Delta \vec{p}_x\right|}{\Delta t} = \frac{2mv\sin\theta}{\Delta t} = \frac{2\times3}{0.2} \times 10 \times \frac{\sqrt{3}}{2}$
= $150\sqrt{3}$

(8) **Answer :** (4)
Solution:
Force on *B* after cutting string
mg = *ma*
a = *g*
force on *A*
3 *mg* – 2 *mg* = 2 *ma*
$a = \frac{g}{2}$

(9) **Answer :** (2)
Solution:
$p = \sqrt{2mk}$

$\frac{p_1}{p_2} = \frac{\sqrt{2m_1k_1}}{\sqrt{2m_2k_2}}$

$\frac{p_1}{p_2} = \frac{3}{2}$

(10) Answer : (3)
Hint:

Work done by a variable force $W = \int \vec{F} \cdot d\vec{s}$

Solution:

$dW = x\,dx + y^2\,dy$

$W = \int dW = \int_1^{-3} x\,dx + \int_2^4 y^2\,dy$

$= \left[\frac{x^2}{2}\right]_1^{-3} + \left[\frac{y^3}{3}\right]_2^4$

$= \frac{68}{3}$ J

(11) Answer : (3)
Solution:

$mk^2 = \frac{ml^2}{12}$

$k = \frac{l}{2\sqrt{3}}$

(12) Answer : (1)
Solution:

$\frac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1+m_2} = \vec{v}_{com}$

$\therefore \quad \vec{v}_{com} = \frac{5\left(2\hat{i}+4\hat{j}\right)+1\times 6\hat{j}}{5+1} = \frac{10\hat{i}+26\hat{j}}{6}$

$= \left(\frac{5}{3}\hat{i} + \frac{13}{3}\hat{j}\right)$ m/s

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(13) Answer : (2)
Solution:

$\tau = I.\alpha \quad \Rightarrow \quad \alpha = \frac{\tau}{I}$

$\alpha = \frac{20}{5} = 4$ rad/s^2

(14) Answer : (2)
Solution:

$\frac{dA}{dt} = \frac{L}{2m}$

(15) Answer : (1)
Solution:

In a satellite, due to free fall, weightlessness is experienced. Kinetic energy of a satellite is positive.

(16) Answer : (4)
Solution:

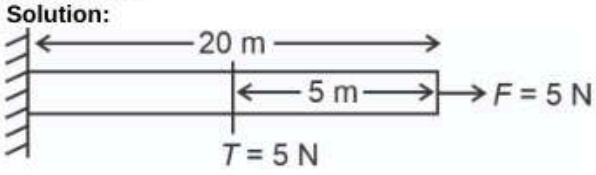
$F_1 = \frac{GM_1M_2}{r^2}$

$F_2 = \frac{G4M_14M_2}{(2r)^2}$

$F_2 = 4F_1$

% increase $\Rightarrow$ $\frac{F_2-F_1}{F_1} \times 100 = 300\%$

(17) Answer : (2)
Solution:



Tension is equal to applied force (within elastic limit) and same throughout in case of massless string.

(18) **Answer :** (1)
Hint:

Average power $P_{avg} = \frac{\text{Total work done}}{\text{Time taken}}$

Solution:

$P_{avg} = \frac{mgh}{t} = \frac{200\times10\times20}{20} = 2$ kW

(19) **Answer :** (3)
Solution:
9 VSD = 6 MSD
⇒ 1 VSD = $\frac{2}{3}$ MSD
⇒ 1 VSD = $\frac{2}{3}$ (0.01 mm)
and L.C. = 1 MSD – 1 VSD
⇒ L.C. = 0.01 mm – $\frac{2}{3}$ (0.01) mm
= $\frac{0.01}{3}$ mm = 0.0033 mm

(20) **Answer :** (4)
Hint:

$X = \frac{A^aB^b}{C^c}$

⇒ $\frac{\Delta x}{x}\% = a\frac{\Delta A}{A}\% + b\frac{\Delta B}{B}\% + c\frac{\Delta C}{C}\%$

Solution:

$k = \frac{P^2}{2m}$

$\frac{\Delta K}{K}\% = 2\frac{\Delta P}{P}\% + \frac{\Delta m}{m}\%$

$\frac{\Delta K}{K}\% = 2\times3 + 2 = 8\%$

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(21) **Answer :** (2)
Hint:
$n_1u_1 = n_2u_2$
Solution:

$10\left[M_1L_1^2T_1^{-2}\right] = n_2\left[M_2L_2^2T_2^{-2}\right]$

$\therefore\ n_2 = 10\left[\frac{1\,\text{kg}}{1\,\text{kg}}\right]^1\left[\frac{1\,\text{m}}{10\,\text{cm}}\right]^2\left[\frac{1\,\text{s}}{30\,\text{s}}\right]^{-2}$

$= 10\times\left[\frac{100\,\text{cm}}{10\,\text{cm}}\right]^2\times\left[\frac{1}{30}\right]^{-2}$

= 10 × 100 × 30 × 30
= 9×10^5 units

(22) **Answer :** (1)
Solution:

Hint : [Force] = $[A]^x\,[v]^y\,[\rho]^z$

Sol. : [Force] = $[MLT^{-2}]$
⇒ $[MLT^{-2}] = [L^2]^x\,[LT^{-1}]^y\,[ML^{-3}]^z$
$2x + y - 3z = 1$
$z = 1$
$-y = -2$
⇒ $x = z = 1$
$y = 2$
$[F] = [\rho]\,[A]\,[v]^2$

(23) **Answer :** (1)
Solution:
d_{rel} = 300 + 200 = 500 m
v_{rel} = (30 + 20) = 50 m s^{-1}

$t = \frac{d_{rel}}{v_{rel}} = \frac{500}{50} = 10\,\text{s}$

(24) **Answer :** (2)
Solution:
V_{rel} = 30 m/s
a_{rel} = 0
$t_{cross} = \frac{60}{30} = 2$ s
$s_1 = \frac{1}{2} \times 10 \times 2^2 = 20$ m
$s_2 = 30 \times 2 - \frac{1}{2} \times 10 \times 2^2 = 40$ m

(25) **Answer :** (2)
Solution:
For circle $a_{Total} = \sqrt{a_C^2 + a_t^2}$
$a_c = \frac{v^2}{R} \Rightarrow$
At $t = 2$, $a_c = \frac{(2\times2)^2}{4} = 4$ m/s^2
$a_t = \frac{dv}{dt} = 2$ m/s^2
$a_{Total} = \sqrt{4^2 + 2^2} = \sqrt{20}$ m/s^2

(26) **Answer :** (2)
Solution:
a. (iii) $x = (10 \times 2) + \left(\frac{1}{2} \times 2 \times 2^2\right)$ = 20 + 4 = 24 m
b. (iv) $y = \frac{1}{2} \times 2 \times 2^2 = 4$ m
c. (ii) $\vec{v} = 10\hat{i} + 4\hat{i} + 4\hat{j} = \left(14\hat{i} + 4\hat{j}\right)$ m s^{-1}
$\left|\vec{v}\right| = \sqrt{(14)^2 + 4^2} = 2\sqrt{49+4} = 2\sqrt{53}$ m s^{-1}
d. (i) $\left|\vec{a}\right| = 2\sqrt{2}$ m s^{-2}

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(27) **Answer :** (4)
Solution:
$R = 4\sqrt{h_1 h_2}$
$= 4\sqrt{16 \times 9}$
$= 4\sqrt{144}$
R = 48 m

(28) **Answer :** (4)
Solution:
400-300=30a
a=10/3 m/s^2

(29) **Answer :** (3)
Hint:
F = Number of bullet fired per second × (Change in momentum of one bullet)
Solution:
$F = n \times mv$
$200 = n \times \frac{50}{1000} \times 500$
$n = \frac{200}{25} = 8$ bullets per second

(30) **Answer :** (3)
Hint:
Energy stored in stretched spring $U = \frac{1}{2}kx^2 = \frac{F^2}{2k}$
Solution:
Given $K_A = 3K_B$
$\frac{U_A}{U_B} = \frac{K_B}{K_A}$
$\frac{E}{E_B} = \frac{K_B}{3K_B}$

$E_B = 3E$

(31) Answer : (4)
Solution:

If two equal masses undergo head on elastic collision then there velocities will be interchanged *i.e.* $\overrightarrow{V_A} = \hat{i} + \hat{j}$ and $\overrightarrow{V_B} = 8\hat{i} + 8\hat{j}$

(32) Answer : (1)
Hint:
Use conservation of energy.
Solution:

$U_i + K_i = U_F + K_F,\ K_F = 0,\ K_i = \frac{1}{2}mv^2$

$U_i = \frac{-GMm}{R} = -mgR,\ U_F = \frac{-GMm}{\frac{3R}{2}} = \frac{-2mgR}{3}$

$\Rightarrow \frac{1}{2}mv^2 = \frac{-2mgR}{3} + mgR$

$\Rightarrow \frac{mv^2}{2} = \frac{mgR}{3} \Rightarrow v = \sqrt{\frac{2gR}{3}}$

(33) Answer : (4)
Solution:

$E_i = E = -\frac{GMm}{2r}$

$E_f = -\frac{GMm}{4r}$

$\therefore$ Energy required = $E_f - E_i = \frac{GMm}{2r} - \frac{GMm}{4r} = \frac{GMm}{4r} = \frac{-E}{2}$

(34) Answer : (4)
Solution:

$I = MR^2 + MR^2 = 2MR^2$

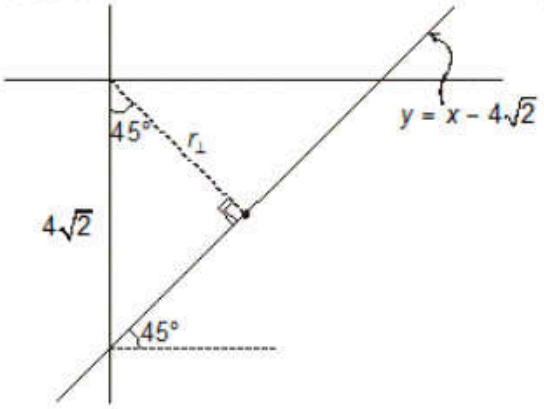
(35) Answer : (2)
Solution:

By conservation of angular momentum, $I_1\omega_1 = I_2\omega_2$

$\frac{2}{5}MR^2 \times \frac{2\pi}{T_1} = \frac{2}{5}M\left(\frac{R}{\sqrt{3}}\right)^2 \times \frac{2\pi}{T_2}$

$\Rightarrow T_2 = \frac{T_1}{3} = \frac{24}{3}$ = 8 hours

(36) Answer : (1)
Solution:



From figure, $r_\perp = 4\sqrt{2}\cos 45^\circ = 4$

Now, $L = mvr_\perp$ = (0.3)(20)(4) = 24 kg m^2s^{-1}

(37) Answer : (1)
Solution:

Since work done by all the forces equal change in K.E.
⇒ Positive work increases K.E. and vice versa.

(38) Answer : (3)
Solution:

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Since the gravitational force is a conservative force hence speed of object on reaching ground will be 1 : 1.

(39) **Answer :** (4)
Solution:

$x = u \times t + \frac{a_x t^2}{2}$

$= 4 \times 3 + \frac{6}{2}(3)^2 = 12 + 27 = 39$ m

(40) **Answer :** (4)
Hint:

For constant retardation stopping distance $S = \frac{u^2}{2a}$

Solution:

$\frac{S_1}{S_2} = \frac{u_1^2}{u_2^2}$

$\frac{30}{S_2} = \left(\frac{1}{2}\right)^2$

S_2 = 120 m

(41) **Answer :** (4)
Solution:

The reading having minimum least count will be the most precise reading. So 10.001 cm is most precise.

(42) **Answer :** (3)
Solution:

$\vec{v}_A = 8t\hat{i}$ m/s

$\vec{v}_B = 6t\hat{j}$ m/s

$\vec{v}_{AB} = \vec{v}_A - \vec{v}_B = \left(8t\hat{i} - 6t\hat{j}\right)$ m/s

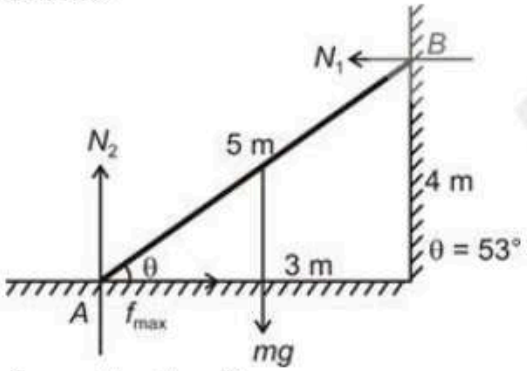
$\left|\vec{v}_{AB}\right| = 10$ m/s

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(43) **Answer :** (3)
Hint:

At equilibrium state of ladder, net force on ladder is zero and net torque acting on it is also zero.

Solution:



$f_{max} = \mu N_2 = N_1$ ...(i)

$N_2 = mg$ ...(ii)

Taking torque about point '*A*'

$mg\frac{5}{2}\cos\theta - N_1 4 = 0$ ...(iii)

On solving (i), (ii) and (iii)

$\mu = \frac{3}{8}$

(44) **Answer :** (1)
Solution:

Escape velocity from the Earth's surface

$v_e = \sqrt{\frac{2GM}{R}}$

$= \sqrt{\frac{2G\rho\frac{4}{3}\pi R^3}{R}}$

$= \sqrt{\frac{8G\rho\pi}{3}R^2}$

$v_e \propto R$ (For same density)

$\frac{v}{v_1} = \frac{R}{4R}$

$v_1 = 4v$

(45) Answer : (2)

Hint:

$v = \frac{dx}{dt}$ and $a = \frac{dv}{dt}$

Solution:

$v = \frac{dx}{dt} = 12 - 3t^2$

$a = \frac{dv}{dt} = -6t$

Now $v = 0 \Rightarrow 12 - 3t^2 = 0 \Rightarrow t = 2\,s$

$\Rightarrow a_{(t = 2\ s)} = -6 \times 2 = -12\ m/s^2$

CHEMISTRY

(46) Answer : (2)

Solution:

Physical Quantity	**– SI unit**
Electric Current	– Ampere
Thermodynamic temperature	– Kelvin
Length	– Metre
Amount of substance	– mole

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(47) Answer : (4)

Solution:

Zeros between two non zero digits are significant.

(48) Answer : (3)

Solution:

Law of Multiple proportion is applicable to those compounds which are made of same elements.

(49) Answer : (4)

Hint:

Average atomic mass $= \frac{\sum \%\ \text{of isotope} \times \text{atomic mass}}{100}$

Solution:

Average atomic mass $= \frac{60 \times 60 + 40 \times 62}{100} = 60.8\ \text{amu}$

(50) Answer : (1)

Solution:

a. 18 g $C_6H_{12}O_6$

Number of oxygen atoms

$= \frac{18}{180} \times 6 \times N_A$

$= 0.6\ N_A$

b. 44.8 L NO_2 gas at STP

Number of oxygen atoms

$= \frac{44.8}{22.4} \times 2 \times N_A = 4\ N_A$

c. 1.225 g $KClO_3$

Number of oxygen atoms

$= \frac{1.225}{122.5} \times 3 \times N_A = 0.03\ N_A$

d. 11.2 L CO gas at STP

Number of oxygen atoms

$= \frac{11.2}{22.4} \times 1 \times N_A = 0.5\ N_A$

(51) **Answer :** (3)
Solution:

1 mole of atomic H = 6.022×10^{23} hydrogen atoms.

(52) **Answer :** (4)
Hint:
Atomic mass of sulphur is 32 u
Solution:
For minimum molecular mass at least one sulphur atom should be present

Mass percentage of an element = $\frac{\text{Mass of an element}}{\text{Molecular mass}} \times 100$

$8 = \frac{32}{\text{Molecular mass}} \times 100$

Molecular mass = 400 u

(53) **Answer :** (2)
Solution:

Element	%	Mole	Simplest Ratio
C	26	2.167	2.167/2.167 = 1
H	4.35	4.35	4.35/2.167 = 2
O	69.65	4.35	4.35/2.167 = 2

Empirical formula $\Rightarrow CH_2O_2$

(54) **Answer :** (2)
Solution:

$A + 2B \rightarrow 3C$

2 2 0

1 0 3

3 moles of c is formed.
1 mole of A is left after the reaction.

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(55) **Answer :** (4)
Hint:

Number of mole(s) $= \frac{\text{Number of molecules}}{N_A}$

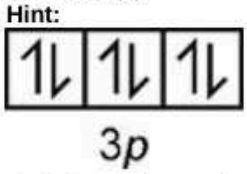
Solution:

Mole of sugar $= \frac{3.01 \times 10^{23}}{6.02 \times 10^{23}} = 0.5$

Mole fraction of sugar $= \frac{0.5}{0.5+24.5} = 0.02$

(56) **Answer :** (3)
Solution:
Cathode rays originate from cathode.
The charge to mass ratio of cathode rays is independent of gas present in cathode ray tube.

(57) **Answer :** (4)
Hint:



Only three electrons have $+\frac{1}{2}$ and other three have $-\frac{1}{2}$ spin quantum number.

Solution:

	Number of electrons
K^+	18
Ca^{2+}	18
S^{2-}	18

These are isoelectronic species
• Shape of orbital is determined by Azimuthal quantum number.

(58) Answer : (4)
Solution:
• CH_3^+ ⇒ 6 + 3 – 1 = 8 electrons
• H_3O^+ ⇒ 3 + 8 – 1 = 10 electrons
• NH_3 ⇒ 7 + 3 = 10 electrons
• H_2O ⇒ 2 + 8 = 10 electrons

(59) Answer : (2)
Hint:
$E = \frac{hc}{\lambda}$
Solution:
• Power of bulb = 100 watt = 100 Js^{-1}
• Energy of one photon, $E = \frac{hc}{\lambda}$
$= \frac{6.6\times10^{-34}\times3\times10^{8}}{400\times10^{-9}}$
$= 4.9 \times 10^{-19}$ J
• Number of photons emitted = $\frac{100\ J}{4.9\times10^{-19}}$
$= 2 \times 10^{20}$

(60) Answer : (1)
Solution:
Diffraction and interference are explained by wave nature of electromagnetic radiations.

(61) Answer : (2)
Solution:

n_1	n_2		Series
1	2, 3...	→	Lyman
4	5, 6...	→	Brackett
3	4, 5...	→	Paschen
5	6, 7...	→	Pfund

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(62) Answer : (2)
Solution:
$\lambda = \frac{h}{mv}$
$= \frac{6.6\times10^{-34}}{20\times10^{-6}\times500}$
$\lambda = 6.6 \times 10^{-32}\text{m}$

(63) Answer : (2)
Solution:

Orbital	Number of radial nodes (n – l–1)
2*p*	2 – 1 – 1 = 0
3*p*	3 – 1 – 1 = 1
3*s*	3 – 0 – 1 = 2
4*s*	4 – 0 – 1 = 3

(64) Answer : (2)
Hint:
The value of m varies from –ℓ to +ℓ
Solution:
For ℓ = 2, m = –2, –1, 0, +1, +2
So for ℓ = 2, m = +3 is not possible

(65) Answer : (3)

Solution:
In SF_6, S has expanded octet.

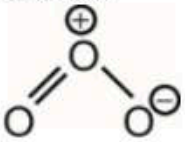
(66) **Answer :** (1)
Solution:
LiCl has highest lattice enthalpy.

(67) **Answer :** (3)
Solution:
Resonance is hypothetical phenomenon which involves delocalisation of pi electrons and resonating structures have same unpaired electrons.
Resonance hybrid is the hybrid of all canonical forms and is most stable.

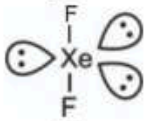
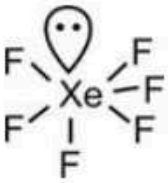
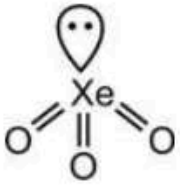
(68) **Answer :** (1)
Solution:

Molecule	Shape
ClF_3	Bent T-shape
XeF_4	Square planar
PCl_5	Trigonal bipyramidal
NH_3	Trigonal pyramidal

(69) **Answer :** (3)
Solution:



(70) **Answer :** (3)
Hint:
Central atom in XeF_6 has six bond pairs and one lone pair.
Solution:

Molecule	Shape	Hybridisation of central atom
XeF_2		sp^3d
XeF_6		sp^3d^3
XeO_3		sp^3

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(71) **Answer :** (1)
Hint:
Species having unpaired electrons are paramagnetic in nature.
Solution:
Electronic configuration of O_2 according to MOT
$O_2 : \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_z^2,$
$\pi 2p_x^2 = \pi 2p_y^2, \pi^* 2p_x^1 = \pi^* 2p_y^1$
Since O_2 has two unpaired electron in π* molecular orbitals it will be paramagnetic in nature.

(72) **Answer :** (2)
Solution:

C_2 has only π bonds

(73) **Answer :** (3)
Hint:
H – bonding within the same molecule is known as intramolecular H – bonding.
Solution:


Involves intermolecular H – bonding.

(74) **Answer :** (2)
Solution:
Equilibrium is dynamic in nature that is both forward and backward processes take place with same rate.

(75) **Answer :** (2)
Hint:
For any gaseous equilibrium
$K_P = K_C(RT)^{\Delta ng}$
Solution:
$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$; $K_P = K_C$
$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$; $K_P = K_C(RT)^{-2}$
$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$; $K_P = K_C(RT)$
$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$; $K_P = K_C(RT)^{-1}$

(76) **Answer :** (3)
Solution:
In an equilibrium, if all the reactants and the products are present in the same phase, it is called homogeneous equilibrium.
$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

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(77) **Answer :** (2)
Solution:
$N_2 + 3H_2 \rightleftharpoons 2NH_3 \quad K_C = K_1$
$NH_3 \rightleftharpoons \frac{1}{2}N_2 + \frac{3}{2}H_2 \quad K_C = \frac{1}{\sqrt{K_1}}$

(78) **Answer :** (3)
Hint:
$\Delta G = \Delta G^\circ + RT \ln Q$
At equilibrium; $\Delta G = 0$ and $Q = K$
Solution:
$\Delta G^\circ = -RT \times 2.303 \log K$
$= -2 \times 300 \times 2.303 \log 10$
$= -1.38$ kcal mol^{-1}

(79) **Answer :** (2)
Solution:
Decreasing pressure results the equilibrium shifts towards more numbers of gaseous mole that will produce more $PCl_3(g)$.

(80) **Answer :** (3)
Solution:
When $\left[\overline{O}H\right]_{SB} < 10^{-7}M$, contribution of OH^- from H_2O is also considered
$H_2O \rightleftharpoons \underset{x}{H^+(aq)} + \underset{10^{-8}+x}{OH^-(aq)}$
$10^{-14} = x\left(10^{-8} + x\right)$
Solving quadratic,
$[H^+] = x = 9.5 \times 10^{-8}$
pH = 7.02

(81) **Answer :** (2)

Solution:
H_2O can act as both acid and base according to Bronsted-Lowry theory.

(82) **Answer :** (1)
Solution:
Salt of strong acid and weak base undergoes cationic hydrolysis.
$NH_4^{\oplus} + H_2O \rightleftharpoons NH_4OH + H^+$

(83) **Answer :** (2)
Hint:
Basic buffer solution (NH_4OH + NH_4Cl)
$pOH = pK_b + \log \frac{[Salt]}{[Base]}$
Solution:
$pOH = 4.74 + \log \frac{\left[\frac{2.5}{75}\right]}{\left[\frac{2.5}{75}\right]}$
pOH = 4.74
pH = 14 – 4.74 = 9.26

(84) **Answer :** (2)
Solution:
$Ag_2CrO_4(s) \rightleftharpoons \underset{2s}{2Ag^+} + \underset{s+0.2\ \sim\ 0.2}{CrO_4^{2-}}$
$\underset{-}{K_2CrO_4} \rightarrow \underset{0.4M}{2K^+} + \underset{0.2\ M}{CrO_4^{2-}}$
$K_{sp} = 1.1 \times 10^{-12} = (2s)^2 \times (0.2)$
$s^2 = \frac{1.1 \times 10^{-12}}{0.2 \times 4}$
$s = 1.17 \times 10^{-6}$ mol/L

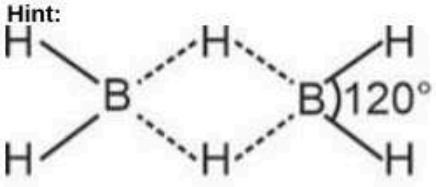
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(85) **Answer :** (3)
Hint:
The relative stability of +1 oxidation state increases for heavier elements (Inert pair effect)

(86) **Answer :** (1)
Solution:
Borax has total 14 'B–O' bonds in its structure.

(87) **Answer :** (1)
Hint:
Boron is a 2nd period member
Solution:
Boron is unable to form BF_6^{3-} ion due to unavailability of *d*-orbitals, it cannot expand its octet.
Boron cannot exceed maximum covalence of 4.

(88) **Answer :** (3)
Hint:



Structure of B_2H_6

Solution:
In B_2H_6, there are two 3c – $2e^-$ bonds and four 2c – $2e^-$ bonds.

(89) **Answer :** (4)
Solution:
Down the group, the size increases and electronegativity decreases and thereby tendency to show catenation decreases.
Lead does not show catenation.

(90) **Answer :** (2)

Solution:
In Buckminsterfullerene, each carbon atom forms three sigma bonds with other three carbon atoms.
All the carbon atoms undergo sp^2 hybridisation in fullerene.

BOTANY

(91) **Answer :** (2)
Solution:
In a life cycle of a fern plant, the correct sequences of stages will be given as follows:
E → D → C → B → A
Spore mother cell undergo sporic meiosis to produce spores
↓
Development of inconspicuous, small, multi-cellular, free living, photosynthetic thalloid gametophyte
↓
Formation of antheridia and archegonia
↓
Fusion of gametes result in the formation of zygote
↓
Development of young embryo takes place within the female gametophyte

(92) **Answer :** (2)
Solution:
In stroma of chloroplast, 70S type of ribosome are present, made up of 30S smaller and 50S larger ribosomal subunit. Light harvesting complex are present in thylakoid. If light becomes unavailable biosynthetic process continues for some time and then stops.

(93) **Answer :** (3)
Hint:
Bacterial envelope consist of glycocalyx, cell wall & plasma membrane.
Solution:
Plasma membrane of bacteria and eukaryotes is similar.
It contains phosphoglycerides with hydrophilic head and hydrophobic tail of saturated hydrocarbons.

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(94) **Answer :** (4)
Hint:
A membraneless organelle which is present in animal cell, but absent in higher plants is centriole.
Solution:
Centrioles have 9 + 0 arrangement of microtubules and they are involved in spindle formation during animal cell division.

(95) **Answer :** (3)
Solution:
ER divides the intracellular space into two distinct compartments.

(96) **Answer :** (4)
Solution:
"*Omnis cellula-e cellula*" explains that all cells arise from pre-existing cells.

(97) **Answer :** (4)
Hint:
These chromosomes appear V-shaped during anaphase.
Solution:
Chromosomes in which centromere is present at the centre are metacentric chromosomes. Metacentric chromosomes contain two equal arms.

(98) **Answer :** (4)
Solution:
Aleuroplast – Stores proteins
Chromoplast – Rich in carotenoids
Elaioplast – Stores fats and oils
Chloroplast – Synthesis of carbohydrates.

(99) **Answer :** (4)
Hint:
Kinetochore is the site where microtubules of spindle attach.
Solution:

On the side of centromere of chromosome, disc shaped structures are present known as kinetochore.

(100) Answer : (2)
Solution:
Adjacent cells in a plant tissue are held together by a thin, sticky, amorphous layer of cementing material which is chiefly made up of calcium pectate.

(101) Answer : (3)
Solution:
Thick and tough glycocalyx is called capsule in some bacteria.

(102) Answer : (4)
Solution:
Axoneme of flagella usually has nine doublets of radially arranged peripheral microtubules.

(103) Answer : (2)
Solution:
Cytotaxonomy is based on cytological information like chromosome number, structure and behaviour.

(104) Answer : (4)
Solution:
Bacteria show nutritional diversity. They may be autotroph (photosynthetic or chemosynthetic) or heterotroph (symbiotic or parasitic or saprophytic)

(105) Answer : (1)
Solution:
Archaebacteria differ from other bacteria in having a different cell wall structure and this feature is responsible for their survival in extreme conditions.

(106) Answer : (1)
Solution:
All single-celled eukaryotes are placed under Protista, but the boundaries of this kingdom are not well defined. Protists reproduce sexually by a process involving zygote formation.

(107) Answer : (1)
Hint:
Deuteromycetes are commonly known as imperfect fungi.
Solution:
Cyanobacteria are characterised by the absence of flagellum throughout the life cycle. Kingdom fungi is classified into various classes on the basis of morphology of the mycelium, mode of spore formation and fruiting bodies.

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(108) Answer : (4)
Hint:
Plasmodium causes malaria.
Solution:
Monera includes prokaryotes (bacteria). Bacteria reproduce by a sort of sexual reproduction by adopting a primitive type of DNA transfer from one bacterium to the other. *Plasmodium* causes malaria, a disease which has a staggering effect on human population. *Saccharomyces* is unicellular. Some hyphae are continuous tubes filled with multinucleated cytoplasm– these are called coenocytic hyphae.

(109) Answer : (3)
Solution:
This organelle is found in almost all eukaryotic cells and is site of ATP synthesis.
Mitochondria are sausage shaped, have their own 70S ribosomes, *i.e,* palade particles and they are viewed after staining with Janus green. Usually their number is high in those cells which have high metabolism.

(110) Answer : (3)
Hint:
At interphase, nucleus contains loose extended & diffused network of nucleoprotein fibres, called chromatin.
Solution:
Term chromatin was given by Flemming and it can be stained by basic dyes. It is composed of DNA and histone proteins. Euchromatin is loosely packed part of chromatin which is transcriptionally active.

(111) Answer : (1)
Solution:
Gametophyte of pteridophytes require cool, damp, shady places to grow, because of this specific restricted requirement and the need of water for fertilisation, the distribution of pteridophytes is limited to narrow geographical regions.

(112) Answer : (4)
Hint:

Red algae reproduce asexually by non-motile spores.
Solution:
Porphyra is a red alga which belongs to the class, Rhodophyceae in which motile cells are absent.

(113) Answer : (1)
Solution:
Mosses along with lichens are the first organisms to colonise on rocks.
Bryophytes in general are of little economic importance but are of great ecological importance.

(114) Answer : (2)
Solution:
Archegonia is female sex organ.
Cycas is dioecious.
Ovules are present in both gymnosperms as well as angiosperms.

(115) Answer : (2)
Solution:
The food is stored as floridean starch in red algae, e.g. *Polysiphonia*.

(116) Answer : (1)
Solution:
Bryophytes are non-vascular terrestrial plants. They are homosporous and produce haploid spores.

(117) Answer : (2)
Solution:
Sequoia is called red wood tree. It is a gymnosperm.

(118) Answer : (1)
Hint:
PS II is found in grana lamellae only.
Solution:
PS I is found on the outer surface while PS II towards inner surface of thylakoids.
Both are involved in non cyclic photophosphorylation but only PS II is associated with release of O_2.

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(119) Answer : (3)
Solution:
To fix a single CO_2 molecule in C_4 plant, 5 ATP molecules are required. Therefore, to form a molecule of hexose sugar (to fix $6CO_2$ molecules) 5 × 6= 30 ATP molecules are required.

(120) Answer : (3)
Solution:
First action spectrum was explained by the experiment performed by T.W. Engelmann.

(121) Answer : (4)
Solution:
Phosphoglycolate -Formed when the most abundant enzyme in the world binds with oxygen
3-phosphoglycerate -Product of the most crucial step of the Calvin cycle
Aspartic acid - Undergoes decarboxylation in bundle sheath cells

(122) Answer : (2)
Solution:
The maize responds to low CO_2 concentrations under high light conditions as compared to tomato.In both plants, water stress leads to reduction in the CO_2 availability. In tomato, RuBisCO is found in mesophyll cells for the carboxylation of RuBP.

(123) Answer : (4)
Solution:
Pigment 'X' is chlorophyll 'a'. It is the chief pigment in photosynthesis process.

(124) Answer : (2)
Solution:
Kranz anatomy is found in C_4 plants, e.g. sugarcane.

(125) Answer : (3)
Solution:
• Sucrose is converted into glucose and fructose by the enzyme invertase.
• Glucose and fructose can easily enter the glycolytic pathway.

(126) Answer : (1)
Solution:

There are four complexes involve in ETC.
Complex I is also known as NADH dehydrogenase.
Complex II is known as succinate dehydrogenase.
Complex III is known as cytochrome bc_1 complex and Complex IV is cytochrome *c* oxidase complex.

(127) Answer : (4)
Solution:
In TCA cycle, substrate for first decarboxylation is oxalosuccinic acid (6C) that gives rise to α-ketoglutaric acid (5C).

(128) Answer : (2)
Solution:
Conversion of pyruvic acid into acetyl CoA is catalyzed by pyruvate dehydrogenase in the presence of Mg^{2+}.

(129) Answer : (4)
Solution:
Complete oxidation of one molecule of acetyl CoA yields 12 ATP.

(130) Answer : (4)
Solution:
Yeast typically poison themselves when the alcohol concentration reaches about 13%. In fermentation, less than 7% of the energy in glucose is released. $NADH + H^+$ is reoxidised to NAD^+ in both lactic acid and alcoholic fermentation. Pyruvic acid is reduced to lactic acid by homofermentative lactic acid bacteria.

(131) Answer : (3)
Solution:
Citric acid – 6 C, Succinyl CoA – 4 C, Oxaloacetic acid – 4 C.

(132) Answer : (4)
Solution:
Oxygen acts as the final hydrogen acceptor in ETS of mitochondria.

(133) Answer : (4)
Solution:
In unicellular algae, reproduction is synonymous with growth.

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(134) Answer : (2)
Hint:
Fungi which show only vegetative phase are known as imperfect fungi.
Solution:
Deuteromycetes is commonly known as imperfect fungi. *Albugo*, is a parasitic fungi on mustard. *Neurospora* is used extensively in biochemical and genetic work. *Ustilago* causes smut disease.

(135) Answer : (3)
Solution:
The small circular DNA outside the genomic DNA is plasmid. It plays an important role in making such bacteria resistant against antibiotics and is also used to monitor bacterial transformation with foreign DNA.

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(136) Answer : (2)
Solution:
$CO_2$and H_2O in presence of carbonic anhydrase reacts to form 600,000 molecules of H_2CO_3 per second.

(137) Answer : (2)
Solution:
Peptide bond is formed when carboxyl group of one amino acid reacts with amino group of next amino acid with the elimination of a water moiety. Glycosidic bond is also formed by dehydration.

(138) Answer : (3)
Solution:
As there are 20 types of amino acids, a protein is a heteropolymer.

(139) Answer : (4)
Solution:
Lecithin is an example of a phospholipid.

(140) Answer : (1)

Solution:

Uridine is a nucleoside. It is formed by linking together of:

• A nitrogenous base (Uracil) – which itself contains one heterocyclic ring.

• A pentose sugar; ribose also contains a heterocyclic ring.

Guanine is a purine with two heterocyclic rings, but it is a single base *i.e.* not formed by a dehydration process.

Cholesterol contains homocyclic rings.

Cytosine is a pyrimidine, which has only one heterocyclic ring, and it exists as a base *i.e.* not formed through dehydration reaction.

(141) Answer : (2)

Solution:

A polysaccharide made of 458 glucose molecules have formed by 457 glycosidic bonds. When completely hydrolyzed, each glycosidic bond is broken by using one water molecule. So 457 water molecules will be consumed in the reaction.

(142) Answer : (4)

Solution:

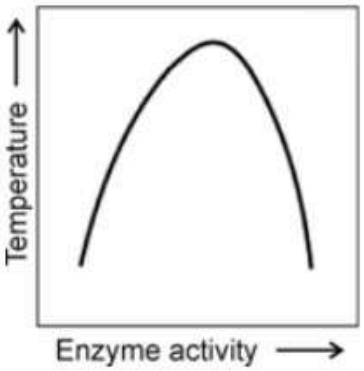
Hydrolases are classified under EC number 3 in the Enzyme Commission (EC) number system.

Maltose is a disaccharide that can be broken into 2 glucose molecules upon hydrolysis.

Rest mentioned sugars are monosaccharides.

(143) Answer : (1)

Solution:



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(144) Answer : (2)

Solution:

A peptide linkage is formed by the removal of a water molecule when two amino acids are joined together.

(145) Answer : (1)

Solution:

When the inhibitor closely resembles the substrate in its molecular structure and inhibits the activity of the enzyme, it is known as competitive inhibitor. Enzyme succinic dehydrogenase is inhibited by malonate which resembles succinate in its structure.

(146) Answer : (4)

Solution:

Catalytic activity is lost when the co-factor is removed from the enzyme which testifies that they play a crucial role in the catalytic activity of the enzyme.

(147) Answer : (1)

Solution:

Primary structure of protein gives the positional information of amino acids in a polypeptide *i.e.*, which is the first, which is the second, and so on.

(148) Answer : (2)

Solution:

Simple sugars, amino acids and nucleotides are present in filtrate.

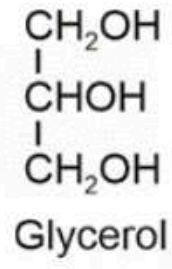
(149) Answer : (3)

Solution:

Antibodies fight against infectious agents. In human body, five kinds of antibodies are found (IgA, IgG, IgM, IgD and IgE).

(150) Answer : (3)

Solution:



Glycerol

Arachidonic acid is unsaturated while, palmitic acid is a saturated fatty acid.

(151) Answer : (4)

Hint:

Identify a layer absent in alveolar capillaries.

Solution:

The diffusion membrane is made up of three layers namely, the thin squamous epithelium of alveoli, the endothelium of alveolar capillaries and the basement substance in between them.

(152) Answer : (4)

Hint:

Affinity of Hb for O_2 is inversely proportional to P_{50} value.

Solution:

Shift of curve B towards right indicates increase in P_{50} value and thus decreased affinity of haemoglobin for oxygen, leading to greater dissociation of oxygen from hemoglobin.

(153) Answer : (2)

Hint:

In alveoli, pO_2 is 104 mmHg and pCO_2 is 40 mmHg.

Solution:

In alveoli of lungs, pO_2 is high but pCO_2 is low in comparison to venous blood. Respiratory rhythm centre regulates normal respiratory rate and is found in medulla oblongata.

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(154) Answer : (1)

Solution:

Decrease in activity of pneumotaxic centre or switch off centre for breathing increases inspiration and expiration time by inhibitory respiratory rhythm centre in medulla. So, decreases breathing rate.

(155) Answer : (3)

Hint:

Pneumotaxic centre moderates function of respiratory rhythm center.

Solution:

Respiratory rhythm centre is situated in medulla to regulate normal breathing rate. Pons varolii contains pneumotaxic centre/switch off centre for breathing. pCO_2 and pH are measured by receptors in chemosensitive area in medulla.

(156) Answer : (1)

Solution:

Increase in pCO_2 and hydrogen ion concentration activates chemosensitive area in medulla to increase breathing rate.

(157) Answer : (1)

Solution:

Total lung capacity (TLC) is the total volume of air accomodated in the lungs at the end of forced inspiration.
Inspiratory capacity is the total volume of air inspired after a normal expiration. Thus, it includes TV (Tidal volume) and IRV (Inspiratory reserve volume).
Vital capacity is the maximum volume of air a person can breathe after a forceful expiration or maximum volume of air a person can expire after a forceful inspiration.
FRC is the volume of air that remains in the lungs after normal expiration.

(158) Answer : (1)

Solution:

7% CO_2 is transported in dissolved form and 70% in the form of HCO_3^- ions through the blood plasma

(159) Answer : (2)

Solution:

Respiration involves the following steps:
(i) Breathing or pulmonary ventilation by which atmospheric air is drawn in and CO_2 rich alveolar air is released out.
(ii) Diffusion of gases (O_2 and CO_2) across alveolar membrane.

(iii) Transport of gases by the blood.
(iv) Diffusion of O_2 and CO_2 between blood and tissues.
(v) Utilisation of O_2 by the cells for catabolic reactions and resultant release of CO_2.

(160) Answer : (4)
Solution:
During normal quiet inspiration, intra-pulmonary pressure is less than the atmospheric pressure.

(161) Answer : (4)
Solution:
A loose connective tissue consists of different types of fibres i.e., white collagen fibres and yellow elastic fibres.

(162) Answer : (4)
Solution:
Stratified epithelium has more than one layer. Due to the presence of mucus secreting goblet cells, columnar epithelium is also known as simple columnar glandular epithelium.

(163) Answer : (3)
Solution:
Muscle fibres contract in response to stimulation then relax and return to their uncontracted state in a coordinated fashion. Their action moves the body to adjust to the changes in the environment and to maintain the positions of the various parts of the body.

(164) Answer : (2)
Solution:
The first pair of wings arises from mesothorax called forewings/mesothoracic wings/tegmina. These are opaque, dark and leathery. The second pair of wings arises from metathorax called hindwings or metathoracic wings. The hind wings are transparent, membranous and are used in flight.

(165) Answer : (4)
Hint:
All female reproductive organs are confined to abdomen.
Solution:
A pair of yellow-coloured ovaries lie laterally in 2nd to 6th abdominal segments, one on either side of hindgut, and embedded in fat bodies.

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(166) Answer : (4)
Solution:
Frogs respire on land and in the water by two different methods. In water, skin acts as aquatic respiratory organ (cutaneous respiration).During aestivation and hibernation gaseous exchange takes place through skin.

(167) Answer : (1)
Solution:
Each kidney is composed of several structural and functional units called nephrons. Sperms are released from testes into kidneys and then into cloaca.

(168) Answer : (3)
Solution:
In *Rana tigrina*, vena cava are veins that carry deoxygenated blood from the body to the heart. These veins open into the sinus venosus, a triangular chamber that acts as a collecting space before passing the blood into the right atrium.

(169) Answer : (4)
Solution:
Metameric segmentation is a characteristic of phylum-annelida, arthropoda and chordata.

(170) Answer : (1)
Solution:
Arthropoda (segmented), Ctenophora (diploblastic), Platyhelminthes (acoelomate).

(171) Answer : (3)
Solution:
Annelids like *Nereis* possess parapodia, which helps in swimming. The body of arthropods is covered by chitinous exoskeleton.

(172) Answer : (2)
Solution:
Obelia and *Physalia* belong to the phylum Coelenterata.

(173) Answer : (1)
Solution:

Taenia – Hooks and suckers
Ctenoplana – Comb plates
Periplaneta – Malpighian tubules
Octopus – Visceral hump

(174) Answer : (3)
Solution:
A body cavity where mesoderm is present as scattered pouches in between ectoderm and endoderm is called pseudocoelom and such organisms are called pseudocoelomates *e.g.* Aschelminthes (*Ascaris*).
Pleurobrachia and *Spongilla* are acoelomates. *Nereis* is a coelomate non-chordate.

(175) Answer : (2)
Solution:

Myxine	– Respiration through 6-15 pairs of gill-slits
Carcharodon	– Absence of air bladder
Calotes	– Tetrapod with epidermal scales
Pterophyllum	– 4 pairs of gills with gill covers

(176) Answer : (2)
Solution:
Lamprey has circular mouth without jaws. They are ectoparasites on some fishes. They lack paired fins and scales. Magur (bony fish) has paired fins and scales.

(177) Answer : (3)
Solution:
Reptiles and birds both are mostly oviparous.

(178) Answer : (3)
Hint:
Pelvic claspers are found in cartilaginous fishes.
Solution:
Trygon also known as sting ray is a cartilaginous fish. It has a poisonous sting while electric organ is present in *Torpedo*.
Swim bladder is found in bony fishes.

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(179) Answer : (2)
Solution:
Hyla is an amphibian that performs external fertilisation.
Hemidactylus and *Testudo* are reptiles that are oviparous.
Betta is a bony fish. Bony fishes are mostly oviparous and show direct development.
Columba is a bird. Birds are oviparous.
Camelus, *Canis* and *Ornithorhynchus* are mammals. Most of the mammals are viviparous and show internal fertilization with direct development.
Ornithorhynchus is oviparous.

(180) Answer : (3)
Solution:
In urochordates (*Doliolum*), it is present only in the larval tail while in cephalochordates (Lancelet) and cartilaginous fishes(shark), it is present throughout their life.

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